

The Contemporary Artist's Guide to Industrial Displays

Contents

1	The Contemporary Artist's Guide to Industrial Displays	3
1.1	1. Executive Summary	4
1.2	2. What Artists Actually Need from Displays	5
1.3	3. The Four Product Classes	6
1.3.1	3.1 Dumb Industrial Touch Monitors	7
1.3.2	3.2 Industrial Panel PCs	7
1.3.3	3.3 Android Signage Displays	8
1.3.4	3.4 OEM / Semi-Barebone Systems	9
1.4	4. Recommended Buying Strategy	9
1.5	5. Shortlist: Best Current Units	10
1.5.1	5.1 Elo 2294L — 21.5" open-frame touch monitor studio standard candidate	10
1.5.2	5.2 Faytech FT215TMCAPOFHBOB — 21.5" open-frame, high-brightness	11
1.5.3	5.3 Mimo M21580C-OF — 21.5" open-frame	11
1.5.4	5.4 OnLogic Tacton TC401 / TN101 — 12.1"/15.6"/21.5" panel PC + matching display-only twin	12
1.5.5	5.5 Cincoze CV/CS series with CDS — modular panel PC, 8.4"–24", to 1800 nits	12
1.5.6	5.6 Geniatech TPC-K3 family — 10.1"–32" Android/Linux ARM panel PCs best hackable Android platform	12
1.5.7	5.7 Waveshare FHD monitor line — 7"–27", the miniature/prototype workhorse	12
1.5.8	5.8 Condensed entries	13
1.6	6. Shortlist: Vendors Worth Contacting	13
1.7	7. Mainboards and SoCs	14
1.7.1	7.1 Rockchip RK3588 / RK3588S — the preferred ARM platform, with open eyes	14
1.7.2	7.2 RK3568 — the budget signage default	15
1.7.3	7.3 x86 — the longevity anchor	15
1.7.4	7.4 External compute — the studio menu	15

1.7.5	7.5 Boards with eDP/LVDS for custom builds	15
1.8	8. LCD Panels, Touch Overlays, and Controller Boards	16
1.8.1	8.1 The panel supply chain has consolidated — plan accordingly	16
1.8.2	8.2 Which sizes survive ten years	16
1.8.3	8.3 Touch controllers and Linux — the honest hierarchy	17
1.8.4	8.4 HDMI-to-eDP/LVDS controller boards	17
1.9	9. Hackability and Serviceability Scoring	18
1.10	10. Product Comparison Table	18
1.11	11. Supplier Questionnaire Results	19
1.12	12. Teardown Findings	20
1.13	13. Recommended Studio Configurations	21
1.13.1	13.1 Simple Looping Video	21
1.13.2	13.2 Interactive Touch Artwork	21
1.13.3	13.3 Generative Real-Time Video	21
1.13.4	13.4 AI / Vision-Enabled Installation	22
1.13.5	13.5 Long-Term Museum Installation	22
1.13.6	13.6 Miniature / Diorama Displays (real-time cinema class)	23
1.14	14. Risks and Failure Modes	23
1.15	15. Final Recommendations	24
1.16	Appendix A: Product Pages and Datasheets	25
1.17	Appendix B: Firmware, Drivers, and Software	26
1.18	Appendix C: Supplier Correspondence Log	26
1.19	Appendix D: Teardown / Inspection Checklist and Testing Protocol	27
1.20	Appendix E: Glossary	27
2	Industrial Panel PCs for Media-Art Installations — Research Report	28
2.1	0. Executive summary / top-line recommendation	28
2.2	1. x86 Panel PCs — Vendor-by-Vendor	29
2.2.1	Advantech (PPC series)	29
2.2.2	AAEON (OMNI series)	30
2.2.3	OnLogic (Tacton line)	31
2.2.4	Cincoze (CV-series, CS-series)	31
2.2.5	Teguar	32
2.2.6	Winmate	33
2.2.7	DFI	33
2.2.8	IBASE	34
2.2.9	Avalue	34
2.2.10	Axiomtek (GOT series)	35
2.2.11	Kontron (FlatClient / KPanel)	35
2.2.12	ASUS IoT	36
2.2.13	Shuttle	37
2.3	2. ARM Panel PCs — Vendor-by-Vendor	37
2.3.1	Forlinx	37
2.3.2	Firefly	38
2.3.3	Advantech (ARM line — PPC-100 series)	38

2.3.4	AAEON (ARM line)	38
2.3.5	Boardcon	39
2.3.6	Geniatech — broadest RK3568/RK3588 size range found	39
2.3.7	Bonus finds (not on original list, relevant)	39
2.3.8	RK3588 Linux/GPU maturity — critical findings for the buying guide	40
2.4	3. Modular "Replaceable Compute" Display Architectures: OPS and SDM	41
2.4.1	OPS (Open Pluggable Specification)	41
2.4.2	Intel SDM (Smart Display Module) / SDM-S / SDM-L	42
2.5	4. Product Lifecycle & Warranty — Cross-Vendor Comparison	43
2.6	5. Real-World Linux Evidence	44
2.6.1	PCAP touch = standard USB HID — the clearest finding of this research	44
2.6.2	N100/N97/Alder-Lake-N specific Linux gotchas	45
2.6.3	Media-art / TouchDesigner / openFrameworks community hardware reality	45
2.7	6. GPU/Real-Time Graphics Comparison: N100-class vs Core-class vs RK3588	46
2.8	7. Sources Index (key URLs by category)	46
2.9	Evidence quality legend	47

1 The Contemporary Artist's Guide to Industrial Displays

Selecting, modifying, and maintaining commercial and industrial displays for long-lived media artworks

Prepared for the studio of Jennifer and Kevin McCoy · July 2026

Scope: Industrial HDMI/USB touch monitors, industrial panel PCs (x86 and ARM), Android signage displays, and OEM/semi-barebone display systems, in sizes from 7" to 43"+. Operating-system scope covers both Android and Ubuntu/Linux, including mainline kernel status for Rockchip SoCs, kiosk-mode Linux configurations, and USB HID touch behavior across platforms.

Evidence labels used throughout: [datasheet] directly verified vendor PDF/spec · [vendor page] vendor web claim · [supplier claim] unverified factory statement · [teardown] physical/photographic evidence · [forum] community/user report · [kernel] Linux kernel source or mailing list · [inferred] reasonable conclusion from adjacent evidence · [unverified] flag for direct confirmation before purchase. Any recommendation based only on an Alibaba listing is marked tentative.

1.1 1. Executive Summary

The question this guide answers:

"Can I buy this display today, run my own software reliably, repair it later, and still keep the artwork serviceable in ten years?"

The short answer: yes — but almost never by buying an integrated Android display. The evidence gathered here (60+ vendors and products surveyed, tear-down and kernel-level sources, museum conservation literature) points consistently to one architecture:

Standardize on "dumb" industrial HDMI + USB-HID touch monitors, and keep the compute external, replaceable, and boring.

Key findings:

1. The dumb-monitor-plus-external-compute architecture won on the evidence (Hypothesis 4, confirmed). A PCAP touch monitor with HDMI input presents itself to any computer — Mac, Linux, Raspberry Pi — as a standard USB HID device, driverless on any Linux kernel ≥ 3.8 [vendor page: Elo support KB]. The display and the compute fail and obsolesce on different timelines; this architecture lets you replace only what breaks.
2. Elo Touch and Faytech open-frame families are the strongest multi-size platforms. Elo's open-frame line spans 7"–42.5" with vendor-confirmed driverless Linux HID touch, published MTBF, and 3–5 year warranties; Faytech spans 7"–43" with published pricing and component-level repairability marketing. No single vendor wins every axis — a two-vendor strategy is recommended (§4, §15).
3. Android signage displays are commodity assemblies on a small set of Rockchip OEM boards (Hypotheses 1–2, supported). Because Rockchip's MaskROM recovery is burned into the silicon, any RK3288/RK3399/RK3566/RK3568/RK3588 unit can always be reflashed with open tools (rkdeveloptool) — but board-specific device trees, panel timings, and touch drivers mean recoverable $\neq$ rehostable. HDMI input remains rare even on 2026 flagship signage hardware [teardown: Rikomagic DS08 review, CNX Software 2026]. Use this class only as a sealed video appliance or for teardown/experimentation.
4. On the Linux side, RK3588 mainline support is real and improving fast but incomplete (eDP and hardware encode lag; Vulkan windowed rendering unreliable), and the most popular community Ubuntu build for Rockchip — Joshua Riek's ubuntu-rockchip — was archived on April 29, 2026 after maintainer burnout [confirmed: github.com/Joshua-Riek/ubuntu-rockchip; jeffgeerling.com]. This is a concrete, already-realized warning: a ten-year plan cannot depend on volunteer distro tooling for Chinese ARM SoCs.

5. The most future-proof compute today is unglamorous: Raspberry Pi 5 (official production commitment to at least January 2036 [vendor page: [raspberrypi.com obsolescence statement](https://raspberrypi.com/obsolescence-statement)]), Intel N100/Core mini PCs and used business-class SFF PCs (standard x86 Ubuntu, no special images), and Jetson Orin for AI/vision work (module production through January 2032 [vendor page: developer.nvidia.com/embedded/lifecycle]).
6. 21.5"/1080p is the safest panel size for ten-year availability — triple-sourced across BOE, AUO, and Innolux in multi-generation industrial families, structurally tied to fab glass-cutting economics (§8). 7"–10.1" is safe via the Raspberry Pi ecosystem's gravitational pull on suppliers. 32" at true 1080p is materially less commodity than it looks.
7. What actually fails, in order: power-supply electrolytic capacitors, LED backlight brightness (L70 half-life math in §14), touch controllers (usually via power/EMI, not glass wear), and HDMI/EDID handshakes. Every one of these has a cheap, known mitigation — brightness derating, Mean Well-class PSUs, spares purchased at build time, and inline EDID emulators (§14).

Hypothesis verdicts (detailed in §15):

#	Hypothesis	Verdict
1	Alibaba signage units are commodity-part assemblies	Supported [tear
2	Visually different units are rebrands of few OEM platforms	Supported (mode
3	Rockchip units are more hackable than Novatek/MStar appliances	Supported, with th
4	Dumb touch monitor + external compute beats opaque Android signage for longevity	Strongly supporte
5	An ideal multi-size family with consistent touch/mounting/power exists	Partially supporte

1.2 2. What Artists Actually Need from Displays

This section grounds the technical criteria in the studio's actual practice, because "industrial display" marketing is written for factory HMIs and retail kiosks, not for artworks that must outlive their components.

The studio's working reality. The McCoy practice spans database-driven film/TV works with custom playback systems, miniature "real-time cinema" installations combining dioramas, cameras, small displays, and custom software, and generative/blockchain-era works running continuous software loops. Works live in museum collections (MoMA, the Met, the Whitney) under time-based-media conservation regimes — the Met's conservation notes on Every Shot, Every Episode describe a 277-disc custom playback system as a managed conservation object. That is the standard this guide plans for: a

display bought today may still need to boot correctly, in public, in 2036, under someone else's care.

From that practice, the real requirements:

1. Small displays matter as much as large ones. Miniature installations need 7"–11.6" panels that behave like serious displays (HDMI in, proper EDID, VESA or clean mounting, no consumer-tablet sleep behavior) — a size class most industrial vendors treat as an afterthought and most consumer vendors treat as a toy.
2. Custom software is the medium. The studio writes Python, JavaScript/TypeScript, and C/C++; playback systems are bespoke. The display must therefore be OS-agnostic glass: HDMI in, USB HID touch out, no opinion about what drives it. Anything that requires the artwork's logic to live inside the display's own Android build couples the work's survival to a Shenzhen firmware team's attention span.
3. 24/7 gallery duty with unattended recovery. Power cuts happen; museum staff will not reflash a bootloader. Auto power-on after power loss, boot-to-content without input, no sleep/DPMS surprises, and watchdog-supervised software are non-negotiable (§13).
4. Conservation-grade documentation from day one. Museums use identity/iteration reports and equipment-significance ratings (Guggenheim/Phillips 1–3 scale: dedicated vs. obsolete-shared vs. freely substitutable equipment). The studio equivalent: for every piece, record panel model, touch controller, EDID dump, firmware images, and a one-sentence fallback plan — at build time, not during a crisis (§13.5, Appendix D).
5. Repairability over features. Known LCD panel model, known touch controller, openable enclosure, replaceable PSU, commodity internal interfaces (eDP/LVDS), and a vendor who answers technical questions are worth more than brightness, bezels, or bundled CMS software.
6. Honest evidence. Signage marketing is systematically ambiguous — "HDMI" without an in/out qualifier almost always means output-only [listing survey, §3.3]. This guide labels every claim, and the supplier questionnaire (§11, Appendix C) exists to convert "listing says" into "supplier confirmed in writing."

1.3 3. The Four Product Classes

Class	What it is
3.1 Dumb industrial touch monitor	HDMI/DP in + USB HID touch, no computer
3.2 Industrial panel PC	Display + integrated x86/ARM compute
3.3 Android signage display	Alibaba-style advertising machine
3.4 OEM / semi-barebone systems	Enclosure + panel + touch + replaceable compute (incl. OPS/SDM slot)

Class	What it is
-------	------------

1.3.1 3.1 Dumb Industrial Touch Monitors

Verdict: the studio default. These are displays in the traditional sense: a panel, a scaler board, a touch overlay with a USB controller, a power supply, VESA holes. Modern PCAP (projected capacitive) touch enumerates as a standard USB HID multitouch device — Elo's own support documentation states PCAP screens need no driver on any Linux kernel ≥ 3.8 [vendor page: elosupport.elotouch.com article 31649725511447], and the mainline hid-multitouch driver auto-binds to compliant devices [kernel: `drivers/hid/hid-multitouch.c`, in-tree since 2.6.38].

Strengths: total OS freedom; compute replaceable indefinitely; the simplest conservation story ("it is a monitor"). Weaknesses: nearly all vendors — including Elo and Faytech — decline to publish the internal LCD panel model or touch-controller chip on public datasheets [datasheet survey, §5]; auto-power-on-after-outage is documented by almost nobody (Mimo is the notable exception [vendor page]); and the touch-controller silicon can change between date codes of the same SKU (documented for Mimo [forum]). All three gaps are closed by the supplier questionnaire (§11).

Watch for the eGalax/EETI caveat: modern eGalax PCAP works via hid-multitouch, but the controller family has a two-decade history of kernel quirks (HID_QUIRK_NOGET in mainline hid-quirks.c; a new MT_CLS_EGALAX_P80H84 quirk class was still being submitted to the kernel mailing list in February 2026 [kernel: patchew.org]). ILITEK, Elo's in-house TouchPro/COACH, and Goodix-over-USB have cleaner records (§8.3).

1.3.2 3.2 Industrial Panel PCs

Verdict: legitimate for self-contained works — buy modular, buy x86, treat ARM panel PCs as Android appliances unless proven otherwise.

x86 panel PCs (Advantech PPC, AAEON OMNI, OnLogic Tacton, Cincoze CV/CS, Teguar, Winmate, DFI, Axiomtek GOT, Kontron, ASUS IoT, Shuttle) run stock Ubuntu on mainline kernels — the single biggest longevity advantage over ARM. The vendors that matter most here are the ones with split architectures where compute and display are separately replaceable: Cincoze's Convertible Display System, AAEON's OMNI CPU-box + panel kits, DFI's Adaptive Display Platform, and Axiomtek's socketed-LGA GOT models [vendor page]. OnLogic has the best-documented Ubuntu story (Canonical-certified sibling products, Tacton tested on 22.04/24.04, self-serve pricing) [vendor page: ubuntu.com/certified]; DFI and OnLogic are the

only vendors found with dated SKU-level lifecycle commitments (to 2035) rather than generic "10 years" language [vendor page, press].

Two honest caveats. First, GPU: N100/Alder-Lake-N iGPUs are inadequate for shader-heavy real-time work — the TouchDesigner community's consistent verdict is that Intel integrated graphics of any generation is for "noodling," and no industrial panel-PC brand appears anywhere in TouchDesigner/openFrameworks forum history [forum: forum.derivative.ca]. The strongest integrated options are ASUS IoT's APC-125U (Meteor Lake, Arc Xe-LPG) and Shuttle's Arrow Lake-U line. Second, touch: even here, plug-and-play is not universal — OnLogic's own support docs walk through a manual eGalax driver install for some SKUs, warning touch is dead until it's done [vendor page: support.onlogic.com].

ARM panel PCs (Geniatech TPC-K3 10"–32", TouchThink RK3588 line, Advantech's aging RK3399 PPC-1xxW, Firefly-based builds) are cheaper and fanless but inherit every Rockchip Linux caveat in §7 — and many wire touch over I²C to the SoC rather than USB, making touch a BSP-driver question instead of a HID guarantee [vendor docs, inferred].

1.3.3 3.3 Android Signage Displays

Verdict: variable to hostile; use only as a sealed player, or as a donor chassis. The research (Part B–F of the signage stream, §12) confirms the class is built from commodity parts on a small pool of Rockchip OEM boards, sold under dozens of brands. What that means in practice:

- Hackability floor is real: any Rockchip unit answers to MaskROM + rkdeveloptool regardless of brand [docs: github.com/rockchip-linux/rkdeveloptool; forum: Armbian, Kobol wiki]. The ELC WF8382T case — mainline U-Boot built for the Firefly-RK3288 dev board booting on a commercial signage tablet — shows how close these boards sit to public reference designs [forum: forum.armbian.com/topic/36417].
- Hackability ceiling is also real: device-tree panel timings, touch drivers, and DDR blobs are board-specific; generic firmware "difficult to find, most firmware being device-specific" [forum: XDA]. Budget reverse-engineering time or don't start.
- HDMI input is a minority feature even in 2026 (Rikomagic DS08, a flagship RK3588 player reviewed May 2026: two HDMI ports, both outputs [teardown/review: cnx-software.com]), and where present is usually a PiP/capture path through the SoC's video-capture block — latency and HDCP apply — not a monitor-style scaler input [inferred from RK3588 architecture, confirmed by listings].
- No vendor in the entire survey advertises ADB, firmware downloads, or developer access as a feature — except Geniatech, which maintains an actual downloads portal and published board datasheets, and is therefore the standout in this class [vendor page: geniatech.com].

- Android versions stagnate: RK3568 units ship Android 11/12 in 2025–26 with no observed updates; AOSP/non-GMS devices sit outside Google's OTA pipeline entirely [listing survey, structural].

1.3.4 3.4 OEM / Semi-Barebone Systems

Verdict: the most interesting class for a studio standard — two distinct sub-strategies.

(a) OPS/SDM slot displays. Commercial displays with an Open Pluggable Specification slot accept a swappable compute module (80-pin JAE connector, 180×119×30 mm, x86 and ARM modules from 9+ makers; IBASE's IOPS-602 explicitly runs Ubuntu [vendor page]). This is "replaceable compute" as an industry standard — 15 years old, still receiving current silicon (Alder Lake-N modules, Giada) — though NEC/Sharp and Panasonic are migrating flagship lines to Intel's newer, thinner-ecosystem SDM standard, so verify slot type per model [vendor pages, press]. Not hot-swappable despite marketing: power off, wait 30 s, swap in under 5 minutes [vendor doc: SMART install guide].

(b) Build-your-own from components. A known LCD panel (bought by exact model number via Panelook's ecosystem), an HDMI-to-eDP/LVDS controller board (\$12–\$185, RTD2556/RTD2795-class), a PCAP overlay kit with USB controller (\$60–\$300 at 21"–27" from Shenzhen sellers; EETI or ILITEK silicon), a Mean Well PSU, and your own enclosure. Every part is individually documented and individually replaceable — the most conservable architecture on paper — but the supply chain is hobbyist-grade: board-panel pairings must be tested, not assumed (documented cases of spec-matching boards failing on specific panels [forum: badcaps.net; mattmillman.com]), and the classic hardware mistake (backlight driven from the 3.3 V logic rail) physically burns boards [forum: mattmillman.com]. Right for one-off pieces and miniatures; buy tested spares of the exact pairing.

1.4 4. Recommended Buying Strategy

The decision tree:

1. Does the work need touch or display only? Either way, start from Class 1 (dumb monitor). Touch is free to ignore.
2. Does the work need real-time GPU graphics or AI/vision? Choose the compute first (§7.4), then the glass. Never let the display's built-in compute dictate the software stack.
3. Does the enclosure need to be invisible/self-contained? Only then consider a panel PC (modular x86: Cincoze CDS, AAEON OMNI, OnLogic) or an OPS-slot display at 43"+.

4. Is it a miniature? 7"–11.6": Waveshare/Lilliput/GeChic class for prototypes and dioramas; Elo 0764L/1093L or Faytech for museum-bound versions.
5. Is someone selling you an Android advertising machine? Only accept it as (a) a sealed MP4-looping appliance, (b) a Geniatech-class documented platform, or (c) a \$200 teardown donor.

Purchasing rules derived from the evidence:

- Buy the platform, not the unit. Prefer vendors whose family spans sizes with the same touch protocol, power scheme, and mounting (Elo open-frame; Faytech open-frame). One spare per size class serves many works.
- Get it in writing before wiring money. The three facts no public datasheet provides — exact LCD panel model, exact touch-controller chip, auto-power-on behavior — go in the supplier questionnaire (§11). A vendor who won't answer is itself a data point.
- Match the date code. Touch silicon changes within a SKU (Mimo case [forum]). When buying spares, buy them in the same order.
- De-risk the compute separately: Pi 5 (to ≥Jan 2036), used business SFF PCs (bought as identical spares), N100/Core mini PCs, Jetson Orin (modules to Jan 2032). Archive OS images, project code, and udev/kiosk configs the day the piece ships.
- Spend the savings on spares. The delta between an Android AIO and a dumb monitor + Pi buys a spare PSU, a spare touch overlay kit, and an EDID emulator — the parts that actually fail (§14).
- US buying paths: Elo via CDW/B&H/Insight; Faytech, Mimo, On-Logic, Beetratics, Waveshare, Lilliput direct with published pricing; Advantech/AAEON/Axiomtek via Mouser/Newark (mostly quote-only); Alibaba/Shenzhen OEMs only after questionnaire responses, samples first, MOQ 1–2 confirmed.

1.5 5. Shortlist: Best Current Units

Scores are from the §9 rubric (max 60). Full data blocks for the leading units; condensed entries after. Prices are July 2026, street/published, and should be re-verified at order time.

1.5.1 5.1 Elo 2294L — 21.5" open-frame touch monitor studio standard candidate

Vendor: Elo Touch Solutions · URL: elotouch.com/open-frame-touchscreens
 · Price: ~\$573–735 (CDW/resellers) [reseller listing] · MOQ: 1 Screen:
 21.5" 1920×1080, 250 nits [datasheet] · Touch: PCAP 10-pt (TouchPro, spec E330620); SAW/IR variants exist — order PCAP explicitly VESA: 100×100 · OS:

none (HDMI/DP/VGA in, USB touch out) · Power: 12 VDC barrel; cable kits sold as spares [datasheet] Linux: vendor-confirmed driverless HID, kernel ≥3.8 [vendor page] · macOS/Windows: HID plug-and-play [vendor page] Panel model / touch chip: not disclosed (Elo in-house COACH ASIC family) — ask via questionnaire · MTBF: 50,000 h published [datasheet] Family: 0764L 7" / 1064L 10" / 1564L 15.6" / 2094L 19.5" / 2294L / 2494L 23.8" / 2794L 27" / 3263L 32" / 4363L 42.5" Warranty: 3 yr, extendable to 5 · Auto power-on: undocumented — confirm [gap] Evidence quality: datasheet + vendor support KB · Score: 52/60 Recommendation: The reference unit for interactive works 21.5" and up. Weakness: no published pricing or panel transparency; strength: the cleanest Linux touch story in the industry and the widest single-vendor size family.

1.5.2 5.2 Faytech FT215TMCAPOFHB0B — 21.5" open-frame, high-brightness

Vendor: Faytech · URL: faytech.us/touchscreen-monitor/capacitive/open-frame · Price: \$932 published; family 7" \$369 / 11.6" \$628 / 15.6" \$736–861 / 32" \$1,899 / 43" \$3,231 [vendor page] · MOQ: 1 Screen: 21.5" 1920×1080, 1000 nits, 100,000 h backlight [vendor page] · Touch: PCAP 10-pt · VESA: 100×100 · Power: 12 VDC Inputs: HDMI/DP/VGA + USB touch + RS232 on every family SKU Linux: vendor claims hid-multitouch compatibility for current PCAP line [vendor page, generic]; legacy resistive line used EETI eGalax tooling — confirm controller chip per SKU [gap] Serviceability: "each component can be replaced without returning the unit" [vendor page] · Warranty: 2 yr Evidence quality: vendor pages, published pricing; thinner independent Linux evidence than Elo · Score: 47/60 Recommendation: Best published-price 7"–43" family; the high-brightness option; get written confirmation of the touch chip before fleet purchase.

1.5.3 5.3 Mimo M21580C-OF — 21.5" open-frame

Vendor: Mimo Monitors · Price: \$549.99 [vendor page] · Screen: 21.5" FHD, 250 nits · VESA: 75/100 The differentiator: explicit "auto power on after power loss" and "Linux HID support" in vendor copy — the only vendor documenting outage recovery [vendor page] Family: 7" (UM-760), 10.1", 18.5", 21.5", 23.8", 27", 32" — one of the few sources for 18.5" Caveats: touch controller varies by date code (SiS / eGalax / Microchip observed [forum]); macOS needs a \$35 driver; one build-quality complaint [forum] Score: 46/60 · Recommendation: first choice where unattended power recovery must be on paper; pin the date code.

1.5.4 5.4 OnLogic Tacton TC401 / TN101 — 12.1"/15.6"/21.5" panel PC + matching display-only twin

Vendor: OnLogic · Price: from \$987, self-serve configurator (only industrial vendor with real published pricing) [vendor page] CPU: Alder Lake-N → 12th-gen Core i3 · OS: Ubuntu 22.04/24.04 tested; Canonical-certified sibling products [vendor page: ubuntu.com/certified] The clever part: TN101 is the same glass without compute — a panel-PC family and dumb-monitor family in one industrial design Lifecycle: company publishes PCN archive; sibling product guaranteed to 2035 [vendor page] · Touch caveat: some SKUs need manual eGalax driver setup on Linux [vendor page: support.onlogic.com] Score: 47/60 · Recommendation: best x86 panel-PC on documentation and pricing transparency; buy TC401 + TN101 pairs to hedge the architecture decision.

1.5.5 5.5 Cincoze CV/CS series with CDS — modular panel PC, 8.4"–24", to 1800 nits

Vendor: Cincoze · Price: quote [vendor page] · OS: Ubuntu 24.04 listed; RHEL 8 certification on sibling compute [vendor page; catalog.redhat.com] The differentiator: Convertible Display System — display module and compute module independently field-replaceable; the purest "swap the brain, keep the glass" design surveyed Linux evidence: CNX review of DS-1402 compute (same lineage) on Ubuntu 24.04: works well [press/review] · Warranty: 3 yr (2 yr LCD/touch) Score: 49/60 · Recommendation: the panel PC to standardize on if self-contained enclosures become a recurring need.

1.5.6 5.6 Geniatech TPC-K3 family — 10.1"–32" Android/Linux ARM panel PCs best hackable Android platform

Vendor: Geniatech · Price: e.g. TPC1560K4 15.6" ~\$469, TPC2700 27" ~\$549 [listing] · SoC: RK3568 (K3-3568 board; some 32" variants differ — verify) [datasheet] OS: Android 11 + documented Debian/Yocto/Buildroot; official firmware/downloads portal and published board datasheet with ADB/fastboot documentation — unique in the signage class [datasheet: file.geniatech.com/K3_3568_spec] HDMI in: optional add-on board (4096×2160@30, PiP) [datasheet] · RAM: to 8 GB Evidence quality: datasheet-confirmed — the exception in this market · Score: 41/60 Recommendation: if a work truly must be a self-contained ARM Android/Linux unit, this is the only vendor family with the documentation to survive ten years. Ubuntu specifically: unofficial; Debian is the documented path.

1.5.7 5.7 Waveshare FHD monitor line — 7"–27", the miniature/prototype workhorse

Vendor: Waveshare · Price: 21.5" \$159–369; 27" ~\$160; 7"–15.6" \$60–200 [vendor page] Touch: Goodix GT911 behind a custom USB-HID translator

MCU — reliably driverless everywhere; independently reverse-engineered [datasheet; teardown: github.com/pysco68/waveshare-hid] Docs: best step-by-step Linux/Raspberry Pi setup instructions of any vendor, including DPMS-disable guidance [vendor wiki] Caveats: consumer-grade enclosures; USB/USB-C power on many models; no 32"+; long-term availability of exact SKUs unproven Score: 45/60 (long-term suitability capped by build class) · Recommendation: the default for dioramas, prototypes, and studio test rigs; for museum-bound miniatures, step up to Elo 0764L/1093L or Faytech 7"/11.6".

1.5.8 5.8 Condensed entries

Unit	Size(s)	Price	Key fact
Elo 0764L	7" open-frame	~\$250–350 [reseller]	The museum-g
Advantech IDS-3100/3200 series	6.5"–27"	quote	Touch chip do
GeChic T315	15.6"	~\$500	Explicit "design
ViewSonic TD3207	32" open-frame	~\$730–900	IP65/IK08, dua
Lilliput TK-series	7"–21.5"	\$229–500	Vendor-publish
Beetronics TS7M line	7"–27"	\$389–899	9–36 V locking
TouchWo / Raypodo / Shining RK3568 AIOs	10.1"–43"	\$180–600	Cheapest touc
Sunchip AD-C05-RK3566 board	board only	\$64 @ MOQ 2 [listing]	The commodit
Firefly AIO-3588Q	board only	\$429–899	RK3588, eDP

1.6 6. Shortlist: Vendors Worth Contacting

Tier 1 — build the studio standard here:

Vendor	Why
Elo Touch	Driverless-HID certainty, 7"–42.5" family, 50k h MTBF, spare-parts SKUs
Faytech	Published pricing 7"–43", component repairability, 1000-nit options, NY office
OnLogic	Ubuntu certification culture, published PCNs, TC401/TN101 twin architecture, Vermont-bas
Mimo Monitors	Documented auto-power-on + 18.5" coverage

Tier 2 — specific capabilities:

Vendor	Why
Cincoze	CDS modular display/compute split
Geniatech	Only documented hackable Android signage platform; also OPS/ARM modules
Advantech	Documented PenMount touch chain; deep industrial catalog; 10-yr Ubuntu program [vend

Vendor	Why
AAEON	OMNI modular panel kits; Jetson-based NIKY line for AI works
GeChic	24/7-rated 11.6"/15.6" units for miniatures
Axiomtek	Socketed-CPU GOT models; Mouser availability
Hope Industrial	US-made, published pricing, 5-yr warranty, 15"–22" [vendor page - follow up]

Tier 3 — components and experiments: Panelook broker network + Crystal Display Systems (UK; panel identification/replacement service) for panels; Xintai Touch (eBay) / ILITEK-based kit sellers for PCAP overlays; Njytouch / e-qstore for HDMI-to-eDP/LVDS controller boards (pre-flashed to your panel model); Sunchip for commodity Rockchip signage boards; Firefly/Radxa/FriendlyElec for RK3588 compute; Mean Well distributors for PSUs.

Red-flag vendors: any listing that says "quad-core CPU" without naming the SoC, "HDMI" without direction, or Android version older than 11 in 2026; Hushida and YXD returned near-zero verifiable specs across repeated searches [unverified] — treat as opaque until they answer the questionnaire.

1.7 7. Mainboards and SoCs

1.7.1 7.1 Rockchip RK3588 / RK3588S — the preferred ARM platform, with open eyes

Mainline Linux status (mid-2026) [official: Collabora blog/gitlab; cnx-software]:

Subsystem	Status
GPU (Mali-G610)	Panthor driver (not Panfrost) since 6.10; Mesa GLES 3.1 conformant; Vulkan improving
Video decode	H.264/H.265 merged (Collabora, into early 2026); multi-core decode, AV1/VP9 still pending
Video encode	Not yet mainline [inferred - gap]
HDMI out	Basic in 6.13; audio ~6.15/6.16; CEC 6.17
eDP	Weakest output — rides the old rk3399-edp path; native DP lacks audio/HDCP
MIPI-DSI	~6.14
NPU	"Rocket" open driver mainlined July 2025 (RK3588 only) [official: tomeuvizoso.net]
Boot chain	Nearly fully open (U-Boot/TF-A); DDR-training blob still closed

The ecosystem risks, stated plainly: vendor BSP kernels sit at 5.10/6.1 with no public support SLA; Rockchip publishes no Western-style EOL/PCN commitments (the "longevity program" claim circulating online traces to a marketing article, not Rockchip [unverified]); successor chips (RK3668/RK3688) were announced 2025, ~3 years into RK3588's life; and ubuntu-rockchip, the most popular community Ubuntu for these chips, was archived April 29, 2026

[confirmed]. Armbian continues (vendor-kernel and near-mainline branches). Rule: if you deploy RK3588, archive the exact kernel source, U-Boot, images, and build scripts on studio storage the day you ship, and assume you are your own BSP maintainer by year 5.

1.7.2 7.2 RK3568 — the budget signage default

Adequate for kiosk apps, video, and light interactivity; Mali-G52 uses mature mainline Panfrost; Debian 12 boots it by default [docs]. But it's capped around Android 11/12 in shipping products, its NPU has no mainline driver, and it shares all the ecosystem risks above. Fine as a sealed appliance brain; choose RK3588 for anything real-time.

1.7.3 7.3 x86 — the longevity anchor

Standard UEFI x86 means stock Ubuntu LTS, mainline kernels forever, no image hunting, TouchDesigner/oF/Max ecosystems, and certified-hardware programs (OnLogic, Advantech, DFI). Caveats: Alder Lake-N Refresh SKUs (N97/N150/N250/i3-N355) need kernel ≥ 6.11 — use 24.04 HWE [forum, convergent]; N100-class iGPU is a hard ceiling for shader work (§3.2); Intel "Embedded Options Available" ≈ 7 years — check ARK per SKU, don't trust vendor marketing for N-series longevity [vendor page: Intel].

Avoid for hackable systems: Novatek/MStar/SigmaStar appliance chips — encrypted firmware, no public recovery mode, UART-and-service-menu hacking only [forum/teardown]. Amlogic: fine as a media player, Windows-only closed flashing tool, weaker signage-Linux story. Allwinner: FEL recovery exists via sunxi community, but thin evidence in this product category — case-by-case.

1.7.4 7.4 External compute — the studio menu

Compute	Price	Graphics/AI
Raspberry Pi 5 (8/16 GB)	\$130–210	VideoCore VII, GLES 3.1/Vulkan 1.2, dual
Intel N100/N150 mini PC	\$130–250	Quick Sync decode; weak 3D
Used ThinkCentre/OptiPlex/EliteDesk SFF	\$90–250	CPU-gen iGPU
Jetson Orin Nano/NX	\$249+	CUDA/TensorRT, 67–157 TOPS
Core-class mini PC / Mini-ITX + discrete NVIDIA	\$400–1500+	Real GPU headroom
Mac mini M4	\$599+	Excellent under macOS/Metal

1.7.5 7.5 Boards with eDP/LVDS for custom builds

Ranked for documentation and survivability: Firefly AIO-3588Q / Core-3588J (eDP 1.3, HDMI, DP; documented Ubuntu/Debian builds; \$289–899) → Forlinx OK3588-C (native eDP, kernel 6.1 Yocto docs) / Boardcon (Android 14/Debian

12 messaging; quote-only) → Radxa ROCK 5 ITX (usable eDP, best mainline momentum) → Sunchip/Geniatech signage boards (eDP+LVDS+touch headers, \$85–250, Android-first — cheapest, riskiest Linux path). Orange Pi 5 and Khadas Edge2 lack usable eDP/LVDS — disqualified where that's the requirement [vendor docs]. All share chip-level MaskROM/rkdeveloptool recovery.

1.8 8. LCD Panels, Touch Overlays, and Controller Boards

1.8.1 8.1 The panel supply chain has consolidated — plan accordingly

Samsung Display exited LCD in 2020–21; LG Display sold its last LCD fab (Guangzhou) to TCL CSOT, completed March 2025 (~\$1.5 B) [press: TrendForce, OLED-Info]; Sharp is winding down in-house industrial LCD by end-2026 [press: Data Modul]. The real supplier set for the next decade: BOE, AUO, Innolux, CSOT, Tianma — with AUO deliberately pivoting toward industrial/automotive [press].

1.8.2 8.2 Which sizes survive ten years

Panel sizes are baked into fab glass-cut economics (Gen 8.5/8.6 mother glass divides efficiently into 21.5", 23.6–23.8", 27", 31.5") [industry], which is a stronger guarantee than market fashion:

1. 21.5" / 1080p — the safest bet. Triple-sourced, multi-generation industrial families: BOE MV215FHM-N30/40/60/70; AUO G215HAN01.x (five variants, identical mechanicals, marketed for kiosk/signage); Innolux M215HCA [datasheet/listings, multiply confirmed].
2. 18.5" / 1366×768 — AUO G185XW01/G185HAN01, BOE DV185WHM — the industrial HMI staple.
3. 23.8" — solid but thinner multi-sourcing in evidence (Innolux V236BJ1 family confirmed) — verify BOE/AUO equivalents when specifying.
4. 27" — mainstream, thinner formal industrial paper trail.
5. 32" warning: the dominant commodity 32" open-cell (BOE HV320WHB-N80, 6+ distributors) is 1366×768, not 1080p — true FHD at 32" is a much less commodity part [listings].
6. 7"–10.1" — sustained indefinitely by the Raspberry Pi/embedded ecosystem rather than by TV economics.

Rule: specify and record the industrial part number (G-/MV-prefix), which carries an explicit vendor lifecycle commitment (AUO publishes a formal industrial EOL process with 12–24 month notices and last-time-buy windows [vendor page: AU0]) — the consumer SKU of the same glass does not. Single-unit sourcing: Panelook (identify + find compatible alternates) → Crystal Display Systems (identification/replacement service, 3"–82"), AGDisplays drop-in service, Kingtech/DisplayModule (no-MOQ), then eBay/Alibaba for commodity

numbers (\$45–190 bare panels at 18.5”–23.8”). Panelook does not reliably list connector pitch — check the panel datasheet before pairing (0.4 vs 0.5 mm eDP connectors) [docs].

1.8.3 8.3 Touch controllers and Linux — the honest hierarchy

All modern USB PCAP rides `hid-multitouch` → `evdev` → `libinput`. Reliability tiers from kernel and field evidence:

Tier	Controllers
Plug-and-play, clean record	ILITEK (ILI2511/2510/2531; dedicated kernel support since 2013, clean HID V
Works, with history	EETI eGalax modern PCAP — functional via <code>hid-multitouch</code> , but two decades o
Avoid	Legacy eGalax resistive (raw ADC coordinates, <code>xinput_calibrator</code> era); anything

macOS is the weak link for touch: no native touchscreen support; EETI's driver is a kext (hostile territory on Apple Silicon secure boot); userspace translators (Touch-Up, `osxhidtouch`) are the durable fallback [docs/forum]. Another argument for Linux playback machines.

Overlay kits (for custom builds and repairs): single-unit PCAP overlay+USB-controller kits are a real, persistent commodity — Xintai Touch (eBay): 21.5” \$230, 27” \$250 with EETI silicon; AliExpress from ~\$63 at 22”; Western B2B equivalents 3–4× (Obeytouch 23.8” ILITEK, \$950) [marketplace, fetched]. Generic controllers sense through ~1.1–4 mm of glass; document your total glass+air-gap stack so future replacements can actually read through the artwork's glazing [industry]. Air-gap perimeter bonding is the DIY method; optical bonding is not.

Goodix I²C sensors need a \$9–14 I²C-to-USB bridge to become HID — stock two as insurance [marketplace].

1.8.4 8.4 HDMI-to-eDP/LVDS controller boards

The panel-revival path: match panel model on Panelook → buy a pre-flashed board from Njytouch (emails firmware to your panel number) or e-qstore (separate LVDS and eDP kit lines) [vendor/forum: mattmillman.com]. Chips: RTD2556 (HDMI→LVDS/eDP 1080p-class, \$12–32), RTD2795 (HDMI 2.0/DP→4K eDP, \$80–185), RTD2660/2662 “PCB800099” family (LVDS/TTL legacy, open programmer tooling). Known failure modes, all documented in the field: backlight wired to the 3.3 V logic rail overheats the board (fit a separate backlight supply); LVDS vs eDP are electrically incompatible despite similar connectors; firmware/EDID mismatch is the standard “no signal” cause; and a spec-matched board is not guaranteed to drive a spec-matched panel — test the exact pairing you intend to keep as a spare [forum: badcaps.net, mattmillman.com, Hackaday.io].

1.9 9. Hackability and Serviceability Scoring

Each candidate scored 1–5 across twelve categories, total /60. For Class-1 dumb monitors, two categories are interpreted for the class: SoC transparency → scaler/touch-controller transparency, and Android hackability → freedom from firmware dependence (no OS to lock = high score).

Category	1	3	5
SoC / controller transparency	Unknown	Named vaguely	Exact chip
Linux support	None	Possible, unofficial	Official/vendor
Android hackability / firmware independence	Locked appliance	ADB may work	Firmware modifiable
External input	None	Output only	HDMI in + output
Touch compatibility	Unknown/proprietary	Works on Android	USB HID
Internal serviceability	Sealed	Openable, undocumented	Modular, documented
Panel transparency	Unknown	Brand known	Exact model
Compute replaceability	Impossible	Surgery	Designed for replacement
Documentation	Marketing only	Partial	Full manual
Vendor reliability	Unknown reseller	Responsive supplier	Established
Multi-size family	One-off	Similar units	Same platform
Long-term artwork suitability	Risky	Usable w/ caveats	Strong candidate

Bands: 50–60 studio standard candidate · 40–49 good with caveats · 30–39 simple playback only · <30 avoid (unless the enclosure is uniquely valuable).

1.10 10. Product Comparison Table

Unit	Class	Sizes	Price	Linux touch
Elo 2294L (open-frame family)	1	7–42.5"	\$573–735	vendor-confirmed HID
Elo 0764L	1	7"	~\$300	vendor-confirmed HID
Cincoze CV/CS + CDS	2	8.4–24"	quote	Ubuntu listed; some driver in
Advantech IDS-3100/3200	1	6.5–27"	quote	PenMount kernel module
Faytech open-frame	1	7–43"	\$369–3231	claimed HID, thin evidence
OnLogic Tacton TC401/TN101	2	12.1–21.5"	\$987+	Ubuntu tested/certified cultur
Mimo M-series OF	1	7–32"	\$549 (21.5")	vendor-claimed HID
Waveshare FHD line	1	7–27"	\$60–370	best-documented, GT911+H
GeChic T315/T131A	1	11.6–15.6"	~\$500	explicit vendor claim
ViewSonic TD3207	1	15.6–32"	\$730–900	inconsistent docs
Geniatech TPC-K3	2/3	10.1–32"	\$469–549	Debian documented
Lilliput TK	1	7–21.5"	\$229–500	vendor matrix; eGalax
Beetronics TS7M	1	7–27"	\$389–899	mixed; macOS driver req'd

Unit	Class	Sizes	Price	Linux touch
TouchWo/Raypodo/Shining RK AIOs	3	10.1–43"	\$180–600	unknown
Unknown-SoC "quad-core" ad machines	3	any	cheap	unknown

1.11 11. Supplier Questionnaire Results

Status: no supplier correspondence has been sent yet. This section holds the instrument and what public documentation already answers; Appendix C is the log for responses as they arrive (with dates). What the public record already establishes: Elo confirms driverless PCAP HID on Linux ≥ 3.8 but discloses neither panel models nor auto-power-on behavior; Faytech publishes prices and repairability language but not chip identities; Mimo uniquely documents auto power-on; Geniatech is the only signage OEM with published board datasheets and a firmware portal; no Shenzhen AIO vendor advertises ADB or firmware access at all. Everything else below must be asked.

The questionnaire (send before purchasing any unit; file answers with date in Appendix C):

Hello,

I am evaluating this display for a long-term interactive media installation. Could you please confirm the following technical details?

1. What is the exact CPU / SoC model (e.g., RK3568, RK3588)? [skip for monitors]
2. What is the Android version, and do you provide OS updates? [skip for monitors]
3. Can ADB / Developer Options be enabled? Is the bootloader locked?
4. Can you provide the factory firmware image? Is there a USB OTG / MaskROM recovery path for reflashing?
5. What is the exact mainboard model number?
6. What is the exact LCD panel model number (the label on the LCD module)?
7. Does the LCD use eDP, LVDS, MIPI, or another internal interface?
8. What is the touch-controller model (chip, not brand)?
9. Does the touch panel appear as a standard USB HID device? Does it work on Ubuntu Linux and macOS without drivers?
10. Does the unit have HDMI input (not output)? Full-screen, or PiP/capture?
11. Can the display work as a normal HDMI monitor with USB touch?
12. Can the internal Android/compute board be removed or replaced?
13. Do you sell replacement mainboards, power supplies, touch overlays, and controller boards as parts?
14. Does the unit power on automatically after AC power loss, with no button press? Is there any sleep/standby behavior when signal drops?
15. Can you provide internal photos of the unit?

16. Is this model available in other screen sizes on the same mainboard and touch platform?
17. What is the warranty, and how long will this model or compatible replacement parts remain available?
18. For Linux-capable units: which distribution and kernel version do you ship or test, and do you publish the kernel source / device tree?

Questions 14 and 18 extend the base instrument for the studio's unattended-installation and Ubuntu requirements.

1.12 12. Teardown Findings

Status: no studio teardowns performed yet — this section reports published teardown evidence; the inspection protocol for units the studio acquires is Appendix D.

The rebrand economy, documented. Sunchip (Shenzhen) openly operates as the ODM behind many brands; an FCC filing (2ALNC-CX968) names Sunchip as manufacturer of a device retailed under another brand, and a CNX teardown traced a Ugoos-branded box to Sunchip via its MAC OUI [teardown/press]. The Armbian ELC WF8382T thread showed generic Firefly-RK3288 dev-board U-Boot booting a commercial signage tablet unmodified [forum: forum.armbian.com/topic/36417] — these boards are lightly skinned Rockchip reference designs. iFixit's generic "Kingbox" teardown found commodity parts throughout, including memory recycled from a Galaxy S3 [teardown: ifixit.com]. Conclusion: the brand on the bezel identifies the enclosure vendor, not the engineering.

What's inside the monitors. No independent Elo teardown exists (a gap); what is documented is that Elo sells no components and services by whole-unit RMA only [vendor page] — spares strategy: buy whole spare units. Signage-CMS industry sources consistently describe cheap Android ad-machines as consumer TV-box reference boards (\$40–90 BOM) in metal cases [industry blogs, convergent].

Power supplies fail first, and famously. The 2004–2010 capacitor-plague evidence base (Samsung 2x6BW-series, ViewSonic VX-series, Dell, Acer — bulging CapXon electrolytics, flicker-then-fail) remains the clearest teardown literature on display mortality [forum/teardown: badcaps.net, hardforum, s-config]. The physics hasn't changed: heat kills electrolytics; every 10 °C cooler roughly doubles capacitor life (Arrhenius approximation, confirmed by Cornell Dubilier's IEEE paper and Mean Well's own FAQ [datasheet/docs, fetched]). Replacement doctrine: Nichicon/Panasonic/Rubycon 105 °C low-ESR; or better, socket a Mean Well-class external supply from the start.

Generic driver boards: the pairing is the product. Multiple independent

sources document spec-matched controller boards failing on spec-matched panels due to undisclosed firmware/timing differences [forum: badcaps.net; mattmillman.com]. The actionable finding: a "spare" is a tested board+panel pairing, not a part number on a shelf.

1.13 13. Recommended Studio Configurations

Common software base for all Linux configurations (the studio player image): Ubuntu LTS (or Debian stable) · Cage or Weston kiosk-shell compositor (single fullscreen app, no window management; Weston when multiple outputs need app pinning, plus weston-touch-calibrator for persistent touch→output mapping) · systemd unit with Restart=always, WatchdogSec= + sd_notify heartbeat, and RuntimeWatchdogSec= driving the hardware watchdog · overlayroot read-only root filesystem (power-cut immune; updates via commit-and-reboot maintenance window) · DPMS/idle disabled (idle-time=0 in weston.ini; no idle daemon under Cage) · Chromium pieces add --kiosk --noerrdialogs --autoplay-policy=no-user-gesture-required --disable-session-crashed-bubble --incognito [docs/community, \$7 sources]. BIOS/firmware set to power on after AC loss. Archive the image per-artwork.

1.13.1 13.1 Simple Looping Video

Glass: commodity commercial display or Elo/Faytech monitor (no touch needed) · Compute: Raspberry Pi 5, NVMe boot, mpv --loop under Cage or the hardware-decode path via GStreamer · Or, where a sealed single object is required: an RK3568 Android unit used strictly as an appliance (USB-stick MP4 autoplay), accepted as disposable. Inline EDID emulator; brightness at 60–80%. Budget ≈ \$150 compute + display + \$60 emulator + spare PSU.

1.13.2 13.2 Interactive Touch Artwork

Glass: Elo 2294L/2494L (or Faytech high-brightness where sunlight matters) · Compute: Pi 5 for 2D/web interactivity; N100/Core mini PC for heavier logic · Touch arrives as HID; map to output persistently (Weston calibrator or LIBINPUT_CALIBRATION_MATRIX udev rule; on X11, scripted xinput map-to-output) · Verify multi-touch and calibration on Ubuntu and macOS at acceptance testing (Appendix D). Spares: one identical monitor per size class, same date code.

1.13.3 13.3 Generative Real-Time Video

Compute first: Mini-ITX or Core-class mini PC with discrete NVIDIA GPU for TouchDesigner/openFrameworks-class work (the community-standard path

[forum]); Arc Xe-LPG-class iGPU (ASUS APC-125U, Arrow Lake-U) only for moderate loads; RK3588 only if the piece targets GLES 3.1 specifically and the studio accepts BSP stewardship (§7.1) · Glass: any \$5 monitor · Watchdog + auto-login + app-restart supervision are mandatory; GPU driver version pinned and archived.

1.13.4 13.4 AI / Vision-Enabled Installation

Compute: Jetson Orin Nano/NX (CUDA/TensorRT; modules supported to Jan 2032; buy the module + carrier, not the devkit, for museum-bound work [vendor page]) · JetPack version pinned and archived; camera pipeline documented (sensor model, lens, mount) with a spare camera purchased at build time · Glass: per \$5; miniature works pair Orin with the 7"–11.6" tier. Plan one JetPack migration mid-life.

1.13.5 13.5 Long-Term Museum Installation

Everything above, plus the conservation layer — this is where the studio's museum experience (Met TBM, MoMA media conservation, Guggenheim CCBA practice) becomes procedure:

1. Identity vs. iteration documentation (Met model): one report for what the work is; one per installation instance for the equipment actually used.
2. Equipment significance rating (Guggenheim/Phillips 1–3) per component, agreed and written down: is this display dedicated (its look is the work), or substitutable (any 21.5" 1080p panel will do)? This single judgment drives the spares budget.
3. The build-time archive: panel model (from the LCD module label, not the product SKU), touch-controller chip, mainboard/scaler model, EDID binary dump, firmware images, OS image, source code + toolchain, udev/kiosk configs, wiring photos, PSU model, glass-stack thickness, and a one-sentence fallback plan ("if this display becomes unobtainable, acceptable substitute is ____").
4. Spares locker per work: 1–2 identical displays (same date code) or panel+controller tested pairings; 2× PSU; 1× touch overlay kit; EDID emulator; the compute unit ×2. Matters in Media Art's equipment condition report template is the base form [museum source: mattersinmediaart.org].
5. Signal-format risk is separate from hardware risk — record what the source expects (resolution, EDID, HDCP posture), the lesson of works stranded by protocol death rather than hardware death [museum source: Smithsonian TBMA].

1.13.6 13.6 Miniature / Diorama Displays (real-time cinema class)

The 7"–11.6" tier for camera-and-miniature works: prototype on Waveshare 7"/10.1" HDMI-touch panels (GT911→HID, best setup docs, \$60–150) driven by Pi 5; exhibit on Elo 0764L/1093L or Faytech 7"/11.6" (industrial PSU, VESA, warranty). Where the display must be embedded in the model itself: bare panel + RTD-class controller board, pre-flashed by Njytouch to the exact panel — buy the pairing twice, test both (§8.4). Beware USB-C-powered consumer panels' sleep behavior; barrel-DC industrial units with documented no-signal behavior are worth the premium in any unattended diorama.

1.14 14. Risks and Failure Modes

Risk matrix (likelihood × impact for a ten-year installation):

Risk	Likelihood	Impact
PSU electrolytic capacitor failure	High (yrs 3–7)	Medium (repair)
Backlight decay to L70/L50	Certain (math below)	Medium
Vendor discontinues exact SKU	High	Low–High (per
Android firmware abandonment	Certain	High if OS-cou
Community distro abandonment (ARM)	High (precedent: ubuntu-rockchip, 2026)	High
Touch controller quirk/regression	Medium	Medium
HDMI/EDID handshake failure	Medium	Medium (looks
Power-cut boot failure	Medium	High (unattend
LCD image retention / OLED burn-in	Low (LCD, reversible) / High (OLED, permanent)	Low / High
Panel model extinct at repair time	Medium (size-dependent)	High
macOS touch driver breakage	High (kext deprecation)	Medium
HDMI-in on Android units is PiP/capture	High	Medium (laten

The backlight math (years to half-brightness, L50): a 30,000 h budget panel run 24/7 crosses L50 in 3.4 years; 50,000 h in 5.7; 70,000 h industrial in 8.0; 100,000 h in 11.4. At 12 h/day these double [datasheet-derived]. Derating brightness pushes every figure right — the single highest-leverage, zero-cost longevity act.

Mitigations ranked by impact-per-effort: (1) brightness derating; (2) LCD over OLED for static content; (3) ventilated enclosures + cheap logged temperature sensor; (4) industrial panels + name-brand PSUs; (5) spares purchased at build time, not at failure time; (6) inline EDID emulator; (7) content-layer pixel-shift/rotation; (8) simple signal chain with documented power sequencing.

1.15 15. Final Recommendations

Best overall studio standard — the architecture, not a product: industrial HDMI/USB-HID PCAP touch monitors + external replaceable compute + the \$13 studio player image + the \$13.5 conservation archive. Vendor pairing: Elo open-frame where driverless-touch certainty and warranty matter most (21.5"–42.5" public-facing works), Faytech open-frame where published pricing, brightness, and the 7"–43" span matter, with Mimo filling 18.5" and documented auto-power-on needs.

Best dumb touch display: Elo 2294L (21.5", score 52) and its family siblings; runner-up Faytech FT215TMCAPOFHBOB for high-brightness contexts.

Best hackable Android signage unit: Geniatech TPC-K3 family (RK3568, 10.1"–32") — the only signage platform with published board datasheets, a firmware portal, and documented ADB/multi-OS support. Honorable mention at the board level: Sunchip RK3588/RK3566 boards for experiments, via MaskROM/rkdeveloptool.

Best panel PC: OnLogic Tacton TC401/TN101 (Ubuntu documentation, published pricing, display-only twin) for standard duty; Cincoze CDS where modular compute-swap is decisive; ASUS APC-125U / Shuttle Arrow Lake-U where iGPU headroom is needed in one enclosure.

Best low-cost experiment: a \$64 Sunchip AD-C05-RK3566 board (MOQ 2) plus any \$180–250 TouchWo/Raypodo 21.5" AIO as a teardown donor — together they teach the entire signage-market anatomy for under \$350. On the monitor side, Waveshare's 21.5" FHD touch (\$159–369) is the studio-bench workhorse.

Avoid list: unknown-SoC "quad-core" advertising machines; anything with no HDMI input and no ADB and no firmware; Novatek/MStar/SigmaStar-based smart panels for any hackable purpose; OLED for static content unless budgeted as a consumable; Mac mini as a Linux platform (Asahi M4 non-production); sole-sourcing any artwork's survival on a community ARM distro (the ubuntu-rockchip lesson); consumer USB-C-powered panels in unattended installations; 32" designs that silently assume commodity 1080p panels; and any vendor who refuses the §11 questionnaire.

Hypothesis verdicts, in full:

1. Commodity assembly — supported. Teardowns (Kingbox, ELC), FCC records, ODM self-description, and BOM-level industry commentary all agree: panel + touch overlay + Rockchip board + generic PSU + folded metal [teardown/forum/industry].
2. Rebrand hypothesis — supported (moderate–strong). One directly confirmed cross-brand case (Sunchip→Ugoos via MAC OUI), FCC ODM filings, reference-design U-Boot booting commercial units, recurring OEM prefixes across brands. Not yet an exhaustive brand→board map; no side-by-side PCB match published [teardown/forum].

3. Rockchip > appliance chips — supported. Silicon-level MaskROM recovery + open rkdeveloptool + firmware forums versus MStar's encrypted images and service-menu archaeology. Caveat carried through this guide: recoverable ≠ rehostable — device trees and touch drivers stay board-specific [docs/forum].
4. Dumb monitor + external compute wins — strongly supported. HDMI-in remains rare on signage (2026 flagship counter-example confirmed [teardown/review]); Android versions freeze at purchase; museum conservation practice independently arrives at the same decoupling logic. This is the guide's central recommendation.
5. The ideal multi-size family — partially supported. Elo (7"–42.5") and Faytech (7"–43") each come close with consistent touch/power/mounting; no single vendor also delivers panel transparency, auto-power-on documentation, and published pricing at once. The durable move is standardizing on the protocol stack — HDMI + USB-HID PCAP + VESA + 12–24 V DC — which every Tier-1 vendor speaks.

1.16 Appendix A: Product Pages and Datasheets

- Elo open-frame: elotouch.com/open-frame-touchscreens · 2294L datasheet: docs.elotouch.com/Elo_2294L.pdf · Linux PCAP KB: elosupport.elotouch.com (article 31649725511447)
- Faytech open-frame: faytech.us/touchscreen-monitor/capacitive/open-frame · 21.5" HB product page: faytech.us/product/ft215tmcapofhbob...
- Mimo: mimomonitors.com/products/21-5-inch-m21580c-of-open-frame-display
- GeChic: gechic.com/en/t315-touch-monitor · gechic.com/en/t131a-touch-monitor/spec
- Lilliput Pi compatibility: lilliputdirect.com/raspberry_pi_compatible
- Waveshare: waveshare.com/21.5inch-fhd-monitor.htm · GT911 datasheet: files.waveshare.com/wiki/common/GT911_EN_Datasheet.pdf
- Advantech: advantech.com/en/products/ubuntu-lts-os · IDS monitor series pages
- OnLogic Tacton: onlogic.com/store/computers/panel-pc/tacton · Ubuntu certification: ubuntu.com/certified (OnLogic, Advantech, DFI vendor pages)
- Cincoze: cincoze.com (CV/CS, CDS) · RHEL listing: catalog.redhat.com/hardware/system/detail/134917
- Geniatech: shop.geniatech.com · K3-3568 spec: file.geniatech.com/thcdownloads/geniatech/specification · APC3588: file.geniatech.com/download/APC/APC3588_Specification_V0.8_EN.pdf
- Sunchip boards: sunchip-tech.com/products (AD-C05-RK3566, RK3588 board)
- Firefly: en.t-firefly.com (AIO-3588Q) · wiki.t-firefly.com (Ubuntu/Debian build guides)
- Forlinx: forlinx.net/product/rk3588-som-134.html · Radxa ROCK 5 ITX:

radxa.com/products/rock5/5itx

- ViewSonic TD3207, Beetronics, Xenarc, Winmate, AAEON OMNI, Ax-
iomtek ITC241, Kontron FlatClient, ASUS APC-125U, Shuttle P21AR01:
vendor product pages per §5/§6 (full URL set in the research files)
- Panel references: panelook.com (MV215FHM, G215HAN01, V236BJ1,
HV320WHB-N80 family pages) · AUO industrial EOL policy: auo-lcd.com

1.17 Appendix B: Firmware, Drivers, and Software

- Rockchip flashing: github.com/rockchip-linux/rkdeveloptool · MaskROM
guides: wiki.kobol.io/helios64/maskrom, Firefly/Radxa wikis · Windows
RKDevTool via vendor wikis
- Rockchip mainline status: [gitlab.collabora.com/hardware-enablement/rockchip-
3588/notes-for-rockchip-3588](https://gitlab.collabora.com/hardware-enablement/rockchip-3588/notes-for-rockchip-3588) · Panthor: [collabora.com blog](https://collabora.com/blog) · NPU
"Rocket": blog.tomeuvizoso.net
- Armbian: armbian.com (vendor + edge branches) · ubuntu-rockchip
post-mortem: [jeffgeerling.com/blog/2024/popular-rockchip-sbc-distro-
limbo-after-maintainer-burns-out](https://jeffgeerling.com/blog/2024/popular-rockchip-sbc-distro-limbo-after-maintainer-burns-out) · archived repo: [github.com/Joshua-
Riek/ubuntu-rockchip](https://github.com/Joshua-Riek/ubuntu-rockchip)
- Touch on Linux: kernel `hid-multitouch` docs: [docs.kernel.org/input/multi-
touch-protocol.html](https://docs.kernel.org/input/multi-touch-protocol.html) · libinput: wayland.freedesktop.org/libinput · ArchWiki
Multitouch Displays · EETI eGTouch Linux guide (v2.5o PDF) · eGalax
kernel quirk history: Launchpad #625511, patchew.org (Feb 2026 series)
- Kiosk stack: Cage ([hjdskes/cage](https://hjdskes.com)) · Weston kiosk-shell + weston-touch-
calibrator · Ubuntu Frame/Core (10-yr maintenance) · overlayroot · Rasp-
berry Pi OS overlay FS (raspi-config)
- macOS touch fallbacks: github.com/shueber/Touch-Up · github.com/daniel5151/osxhidtouch
- Lifecycle statements: raspberrypi.com (obsolescence statements; Pi 5 ≥
Jan 2036) · developer.nvidia.com/embedded/lifecycle (Orin → Jan 2032)
· Intel ARK "Embedded Options Available" flags
- Controller-board resources: mattmillman.com/info/lcd (RTD board series)
· Hackaday.io "All About Laptop Display Reuse" · Njytouch: njytouch.com
· badcaps.net display forum
- Conservation: mattersinmediaart.org (equipment condition report
template) · tate.org.uk Tate Papers 3 (Laurenson 2005) · [metmu-
seum.org](https://metmu-seum.org) TBM working-group documentation · mediaconservation.io ·
tbma.si.edu/hardware-obsolescence

1.18 Appendix C: Supplier Correspondence Log

Empty as of July 2026 — populate with dated responses to the §11 question-
naire.

Date sent	Vendor	Model	Date answered	Panel model	Touch chip	Auto power-on	Firmware	F
-----------	--------	-------	---------------	-------------	------------	---------------	----------	---

1.19 Appendix D: Teardown / Inspection Checklist and Testing Protocol

Exterior: front/rear photos · port cluster · power input · VESA pattern · ventilation · build quality · cable access. Software (Android units): Android/kernel/build versions · Developer Options? · ADB? · unknown APKs installable? · launcher replaceable? · kiosk mode disable? · video loop on boot? · custom app auto-launch? Interior (photograph everything): mainboard + markings · SoC marking · RAM/flash markings · LCD module sticker (the panel model — the single most valuable photo) · touch controller board · PSU board + capacitor brands · cooling · internal USB headers · eDP/LVDS/MIPI cable · HDMI/input board · whether the compute board removes cleanly. External-compute test: HDMI-in from Mac / Pi / Linux box · USB touch on macOS, Ubuntu, Pi OS · calibration accuracy · multi-touch count · auto power-on after cord pull · boots to last input? · any sleep/power-save on signal loss? · EDID reported (save edid-decode output).

Three-mode testing protocol for serious candidates: Mode 1 — built-in playback: MP4 from USB and internal storage; loop seams; auto-start after power cycle; 24-hour soak; codecs; audio path; burn-in behavior. Mode 2 — custom Android app: APK sideload; ADB; launcher replacement; boot auto-launch; immersive fullscreen; touch latency; network and local-file access. Mode 3 — external compute (weight this most): HDMI input from Mac/Pi/Jetson/Linux; HID touch verification (`libinput debug-events`, `dmesg | grep -i hid`); power-recovery behavior; EDID sanity; brightness/OSD control; multi-day stability soak.

1.20 Appendix E: Glossary

ADB Android Debug Bridge — USB/network shell into Android. AOSP Android Open Source Project (no Google services; no Google OTA pipeline). BSP Board Support Package — vendor's kernel + bootloader fork. Cage/Weston Wayland kiosk compositors. eDP / LVDS / MIPI-DSI internal panel-to-board interfaces (modern / legacy / phone-class respectively; electrically incompatible). EDID display's self-description read over HDMI/DP; emulators pin it. HID USB Human Interface Device class — driverless input. hid-multitouch the Linux kernel driver for HID touch digitizers. L50/L70 backlight hours to 50%/70% of original brightness. MaskROM Rockchip silicon-level recovery mode; always answers. MOQ minimum order quantity. OPS/SDM slot-in replaceable compute standards for commercial displays. Open frame monitor sold as bare metal chassis for integration. Panelook the LCD industry's panel-model database. PCAP projected-capacitive touch. PCN product change notification. rkdevel-

optool open-source Rockchip flasher. RTD2556/2795 Realtek scaler chips on HDMI→eDP/LVDS boards. SoC system-on-chip. VESA mount 75/100/200 mm screw pattern.

Research base: five parallel investigation streams (touch monitors; Android signage; panel PCs; SoCs/boards/compute; panels/components/conservation), 200+ sources consulted July 2026, spot-verified on load-bearing claims (Pi 5 longevity, Jetson lifecycle, ubuntu-rockchip archival, LG Display fab sale). Companion research file: *panel-pc-buying-guide.md*. Evidence labels mark the boundary between what is known and what a supplier must still confirm — keep that boundary honest as the document ages.

2 Industrial Panel PCs for Media-Art Installations — Research Report

Prepared for: technical buying guide, media-art studio requirement: Ubuntu/Linux, 10-year serviceability, real-time graphics (openFrameworks/TouchDesigner-class) Research date: July 2, 2026 Method note: Compiled from 5 parallel research passes (60+ searches, mix of direct page fetch and search-snippet extraction — `web_fetch` was rate-limited for large stretches, so many facts are `[vendor page - via search snippet]` rather than fully-verified `[datasheet]`. Every claim below carries an evidence label; treat `[unverified]`/`[inferred]` items as needing a direct vendor RFQ before purchase.

2.1 0. Executive summary / top-line recommendation

- No single vendor wins on all axes. The studio has to trade off between (a) formal Ubuntu certification, (b) genuinely modular/replaceable compute, (c) explicit long lifecycle commitments, and (d) GPU headroom for real-time graphics. Few vendors score high on all four.
- For serviceability + Linux together, the strongest combination is Cincoze CDS, AAEON OMNI, or DFI's box+panel split architecture — all let you swap the compute module without replacing the glass, which is the real lever for 10-year survival (CPUs go obsolete; touch panels/bezels/mounts don't need to).
- For GPU headroom, none of the classic "panel PC" vendors are good enough for serious TouchDesigner/openFrameworks work out of the box — their Intel N-series/Alder-Lake-N iGPUs are explicitly considered inadequate by the TD community for "intensive graphics." Shuttle's Arrow Lake-U and ASUS IoT's Meteor Lake (Arc Xe-LPG) panel PCs are the strongest iGPU options found; anything requiring real discrete-GPU-class

performance points back to custom Mini-ITX builds or OPS/SDM modules paired to a display, not an off-the-shelf sealed panel PC.

- RK3588 ARM panel PCs are not yet a safe bet for Ubuntu-based real-time graphics: Vulkan windowed rendering isn't reliably supported on RK3588 Ubuntu images as of mid-2026, and the most popular community Ubuntu build for RK3588 (Joshua-Riek/ubuntu-rockchip) was archived/discontinued in April 2026. OpenGL ES 3.1 is the safe, functional path.
- OPS is a legitimate longevity strategy — an open, 15-year-old, multi-vendor standard, with modules shipping in current silicon (Alder Lake-N, some Alder-Lake-P) as of 2025. But several major signage vendors (NEC/Sharp, Panasonic, DynaScan) are visibly migrating new flagship lines to SDM/OMI, so OPS should be treated as "good today, verify per-model going forward," not a permanent guarantee.
- The media-art/TouchDesigner community has essentially no track record with industrial panel PC brands. Their default hardware vocabulary is consumer/prosumer mini-PCs (NUC, Zotac, Beelink) or custom Mini-ITX builds with discrete NVIDIA GPUs — not Advantech/AAEON/Cincoze. The studio would be an early adopter of this category within their peer community.
- Best explicit, dated lifecycle commitments: OnLogic (Factor 101, "guaranteed until 2035") and DFI (EC70A-TGU, "availability until Q3 2035") are the only two vendors found with concrete dated SKU-level guarantees rather than generic "10 years" marketing language.

2.2 1. x86 Panel PCs — Vendor-by-Vendor

2.2.1 Advantech (PPC series)

- Models in range: PPC-3151W/3151SW (15.6" FHD), PPC-415W/PPC-421W (15.6"/21.5", refreshed with 13th Gen Intel Core, 2024-2025), PPC-321SW-ADL_N (21.5", Alder Lake-N), PPC-3120/3120S (12.1", Atom E3940). [vendor page — search snippet]
- CPU: Atom E3900 → Celeron → Core i3/i5/i7 up to 13th Gen; Alder Lake-N on slim models.
- Ubuntu/Linux: Strong, formalized. Advantech runs an "Ubuntu Total Solutions"/"Embedded Ubuntu Service" program offering Ubuntu LTS from 16.04 with a stated 10 years of comprehensive support. Linux touch/GPIO/watchdog drivers published, validated against Ubuntu Desktop 22.04.3 LTS and 20.04.5 LTS for named PPC models. Formal Canonical certification (ubuntu.com/certified) skews toward IoT gateway products (UNO-2271G, EI-52), not confirmed for PPC panel PCs specifically. [vendor page] <https://www.advantech.com/en/products/ubuntu-lts-os/>, <https://ubuntu.com/certified/vendors/Advantech>

- Pricing: Quote-based; distributors Newark/element14, Radwell, Industrial Monitor Direct, ProVantage carry specific SKUs but no USD price surfaced in this pass. [distributor page — unconfirmed pricing]
- Modularity: SODIMM DDR4 socket, mSATA/M.2/2.5" SATA bay serviceable; CPU generally soldered/fixed (not swappable) on standard PPC-3xxx/4xxx. "PPC-600/400 Scalable" series suggests more expansion but not confirmed as swappable-compute.
- Fanless: Yes, predominantly.
- Touch: PCAP standard on W-suffix models; resistive on legacy SKUs.
- VESA: Yes, VESA 75 confirmed on PPC-3151 family.
- Lifecycle: No explicit dated statement for PPC line specifically; general Advantech policy publishes 6-month EOL notice and 5-10yr roadmap guarantee. [vendor page]
- Power: Wide-range DC, 9-32VDC typical.
- Real-world Linux evidence [forum]: Advantech's own FAQ portal has multiple entries on touch driver installation under Ubuntu (including "how to make touch work at the login page"), indicating touch calibration is a known friction point, not universally plug-and-play. An older Embedlynx blog post on the PPC-3120 documents manual `inputattach` + `systemd` setup for a serial resistive touch panel — not representative of modern USB PCAP units but a real gotcha for legacy Advantech hardware. <https://www.advantech.com/en-us/support/details/faq?id=1-19MW6AV>

2.2.2 AAEON (OMNI series)

- Models: OMNI-ADN series (2024-2025) — modular HMI panel PCs, 10.4"-21.5", Intel N50 or N97. OMNI-ADP CPU-BOX Kit uses 12th Gen Core/Celeron. [vendor page] <https://www.aaeon.com/en/p/omni-series-modules-panels>
- Ubuntu/Linux: AAEON's broader box-PC line (BOXER, UP series) ships with "Windows 11 IoT Enterprise or Ubuntu Pro 24.04 LTS" — shows an active company-wide Ubuntu Pro program, but OMNI-specific Ubuntu certification not directly confirmed. [inferred]
- Modularity: Best-in-class — genuine CPU-BOX Kit + separate touch panel module architecture, swap compute independently of display. M.2 3052 B-Key + Nano SIM, M.2 2230 E-Key, 2.5" SATA bay, full-size mPCIe.
- Fanless: Yes.
- Touch: PCAP multi-touch, up to 178° viewing angle, 1920x1080.
- VESA: Yes, confirmed for OMNI-ADN.
- Lifecycle: No OMNI-specific statement; general AAEON policy ~3-5 years (Intel roadmap-aligned), 2-year standard warranty, extendable on request.
- Power: 9-30VDC.
- URLs: <https://www.aaeon.com/en/product/list/modular-hmi-panel-pcs-and-displays>, <https://www.electronics-lab.com/aaeon-launches-new->

omni-adn-hmi-panel-pc-series-ranging-from-10-4-to-21-5/

2.2.3 OnLogic (Tacton line)

- Models: Tacton TC401 (compute-equipped panel PC) in 12.1", 15.6", 21.5"; Tacton TN101 (display-only, no compute) same sizes, from \$987. [vendor page, priced directly] <https://www.onlogic.com/store/computers/panel-pc/tacton/>
- CPU: Alder Lake-N up to 12th Gen Core i3.
- Ubuntu/Linux: Best-documented Ubuntu evidence of any x86 vendor. OnLogic is a Canonical certification partner; sibling products (Helix 330, Helix 401) are formally listed on ubuntu.com/certified for Ubuntu 22.04 LTS. Tacton-specific support docs state Ubuntu 22.04 and 24.04 tested for functionality. Dedicated support article: "Ubuntu Certified Hardware: What It Is And Why It Matters." [vendor page] <https://ubuntu.com/certified/202304-31510/22.04%20LTS>
- Pricing: Genuinely configurable/published on OnLogic's own store — the only vendor in this survey with real self-serve pricing (not quote-only).
- Modularity: M.2 storage with additional expansion slots, DDR5 up to 16GB — moderate, not swap-module-level.
- Fanless: Yes.
- Touch: PCAP and resistive options.
- Lifecycle: No Tacton-specific dated commitment, but company-wide: Factor 101 (a different product) is explicitly "guaranteed until 2035" — one of only two vendors in this whole survey with a concrete dated commitment. Public PCN archive at support.onlogic.com/pcn/.
- Real-world Linux evidence [vendor/support page, verified]: OnLogic's own "Touchscreen Driver Installation (Linux)" support doc explicitly states "Some touchscreens require additional drivers to work on Ubuntu" and walks through a manual eGalax driver install (`setup.sh`, select "USB" mode, reboot) — and warns touch will not work until this is done, so a keyboard/mouse is needed during initial commissioning. This directly contradicts a "PCAP always just works" assumption for at least some OnLogic/Cincoze-family units. <https://support.onlogic.com/support-articles/how-tos/firmware-bios-drivers/touchscreen-driver-installation-linux.md>

2.2.4 Cincoze (CV-series, CS-series)

- Models: CV-100/P1000 & P2000, CS-100/P1000 & P2000 (sunlight-readable, up to 1,800 nits), sizes ~8.4"-24" — covers 10.1"/15.6"/21.5"/23.8" targets.
- CPU: Older gen confirmed (6th Gen Core U-series on CV-100/P2000, Atom E3845 on CS-100/P1000, 10th Gen Atom x6425E on newer compute modules); current-gen (N100/N305/13th) not confirmed for panel line in this pass.

- Ubuntu/Linux: Not explicitly confirmed as "Ubuntu certified" for the panel PC line directly, but Cincoze officially lists support for "Windows 11, Windows 10 LTSC, and Linux through Ubuntu Desktop 24.04 LTS" per spec pages [vendor page]. Cincoze's older CV-P2102 is listed in the Red Hat Ecosystem Catalog as RHEL 8 certified — real vendor-neutral enterprise Linux certification. <https://catalog.redhat.com/hardware/system/detail/134917>
- Modularity: Standout — "Convertible Display System" (CDS) explicitly allows independent on-site replacement of either the display module or the compute module. One of the most genuinely modular architectures surveyed — directly relevant to "10-year serviceability" (swap just the compute when it ages out).
- Fanless: Yes.
- Touch: Multi-point PCAP standard.
- Lifecycle: General "10-15 year industry need" framing only, no dated per-SKU commitment found. 3-year standard warranty (2-year for LCD/touch products).
- Power: 9-48VDC wide range.
- Real-world Linux evidence [blog/review, high credibility]: CNX Software/Linuxium reviewed the Cincoze DS-1402 (same product family/BSP lineage) under Ubuntu 24.04 with Core i9-12900E — "works well... no major issues reported," iGPU and discrete NVIDIA GTX 1630 both correctly detected, all 10 GbE ports worked; one complaint was fan noise under load. <https://www.cnx-software.com/2024/07/08/cincoze-ds-1402-review-ubuntu-24-04-intel-core-i9-12900e-embedded-system/>

2.2.5 Teguar

- Models: TP-5645-22, TP-4845-17/22, TP-5945-22, TP-5045-22, TP-5010-22 ("long life cycle" series), TSP-4845-22 (stainless washdown). Coverage skews 12"-22"; 23.8"/27"/32" not clearly confirmed.
- CPU: J6412 (Elkhart Lake) on TP-4845; 11th Gen Core on TP-5945; older 8th Gen on legacy TP-5645.
- Ubuntu/Linux: Teguar markets TP-5645-22 as "a great solution for most Linux or Windows based applications" — direct but informal (not phrased as formal Ubuntu certification). [vendor page]
- Modularity: Not a swap-module architecture; longevity strategy is component-roadmap discipline (fixed BOM sourced from Intel Embedded Roadmap) rather than field-swappable compute.
- Fanless: Yes, core positioning.
- Lifecycle — best explicit language of any x86 vendor found: "Teguar's panel PC products are powered by quad-core Intel processors and built exclusively with Intel Embedded Roadmap components, ensuring long-term availability, extended lifecycle support, and consistent system performance over 10+ years... your system will remain available for up to 10 years without surprise EOL changes." TP-5010 page adds: "available in

this exact same configuration for the life of your project.” [vendor page, high confidence] <https://teguar.com/tp5010-long-life-cycle-panel-pcs/>

- Warranty: 24-month standard, extendable to 5 years (must be purchased at time of sale). Free lifetime phone/email support regardless of warranty status.

2.2.6 Winmate

- Models — broadest size coverage of any vendor: W22IF7T-CHA3/IPA3 (21.5”, 9th Gen Core), W22IK7T-PMA3/OFA3 (21.5”, 7th Gen), W27IAD7T-PCA1/IT7T-PCA1 (27”, 12th Gen Alder Lake-S, LGA1700 socket), W27IAD7T-GPA1 (27” + discrete RTX option), W32IW3S-PTA3/W32IF7T-PCA3/W32L100-PCA3 (32”), R15ITWS-MHB1-EX/MHC3 (modular M-Series with service access window).
- CPU: Broadest range surveyed — 7th through 12th/14th Gen Core, Tiger Lake on some 21.5” models.
- Ubuntu/Linux: W22IF7T-CHA3/IPA3 spec sheets explicitly list “Linux Ubuntu 18.04 (Optional)” as a factory OS option — dated (18.04 is EOL) but shows Winmate treats Ubuntu as a first-class purchasable configuration, not DIY-only. Winmate’s marketing also confirms PCAP touchscreens implement standard USB HID protocols for driverless plug-and-play on Windows/Linux.
- Modularity: M-Series HMI has an explicit, named service-access window: “Users can upgrade the memory, change the hard disk, or access the PCIe Module... an easy access service window.”
- Fanless: Yes.
- Touch: PCAP on “PCA”-suffix newer models; 5-wire resistive on washdown-rated stainless models (e.g., W32IW3S) — check per-SKU.
- Lifecycle: “Product Longevity Program (PLP)” — maintains stable supply for 5 years. 1-year standard warranty, extendable to 3 years.
- URLs: <https://www.winmate.com/en/Product/W27IAD7T-PCA1>, <https://www.winmate.com/en/Product/W32IF7T-PCA3>

2.2.7 DFI

- Models: KSM-AL (Atom-class, 7”-10.1”), KSM-SD (6th Gen Core), KSM-KH (7th Gen Core, 15”/19”/22”). Legacy KS-156AL (15.6”, Apollo Lake).
- CPU: Notably older generation lineup than most competitors — no 12th/13th gen or N100/N305 confirmed for the KSM panel line in this pass.
- Ubuntu/Linux: DFI has a formal, dedicated Ubuntu partnership — co-branded “DFI x Ubuntu” datasheet, dedicated “Ubuntu Certified Products” tag page. Legacy KS-156AL confirmed running Fedora 24, Ubuntu 16.04, and Yocto Linux. Named certified products on ubuntu.com/certified/vendors/DFI are board/box-PC products, not confirmed KSM panel SKUs specifically.

- Modularity: Strong, explicit — "Adaptive Display Platform" splits box module (CPU/memory/I/O) from panel module (touch+display), conceptually similar to Cincoze CDS/AAEON OMNI.
- Fanless: Yes.
- Lifecycle — tied with OnLogic for most concrete dated commitment found: "DFI guarantees up to 15-year product longevity." Concrete example (different product, same policy): EC70A-TGU (Tiger Lake) — "15-Year CPU life cycle guaranteeing availability until Q3 2035", independently reported by CNX Software. EOL notice issued 6 months prior to discontinuance. [press + vendor page] <https://www.cnx-software.com/2022/12/05/dfi-ec70a-tgu-embedded-computer-long-term-support-2035/>
- Power: 9-36VDC (KSM-KH), 9-48VDC (KSM-AL).

2.2.8 IBASE

- Models: OFP-151-PC (15"), OFP-2100-PC/2101-PC (21"), OFP-W2700 (27", open-frame). Current 2025-2026 catalog spans 27", 21.5", 15.1", 12.1", 10.4", 10.1", 7", 19", 17" — nearly the full target range except 23.8"/32". [vendor catalog PDF] https://www.ibase.com.tw/english/download/Industrial_Panel_PC_Catalog/IBASE__Panel_PC_Catalog_Vol2501.pdf
- CPU: Pentium QC N4200 (Apollo Lake, dated) or optional 7th Gen Core i3/i5/i7 on OFP series; newer catalog CPUs not confirmed in this pass.
- Ubuntu/Linux: OFP series OS support confirmed as "64-bit Windows 10 and Linux Kernel 4+" — generic kernel support, not a named Ubuntu certification. IBASE's separate IOPS-602 OPS module explicitly runs both Windows 10 and Linux Ubuntu (see OPS section below) — the strongest direct Ubuntu evidence for IBASE overall.
- Modularity: More integrated/sealed than DFI/Cincoze/AAEON — mini PCI-E slot for half-size mSATA gives some storage serviceability; RAM appears onboard (not confirmed SODIMM).
- Lifecycle: No explicit statement found. 2-year standard warranty, extendable to 5 years.

2.2.9 Avalue

- Models: SPC-1533-B1 (15"), SPC-2133/2133-B1 (21.5"), SPC-10W35 (10.1").
- CPU: Celeron J3455 (Apollo Lake, dated) on SPC-2133/1533; J6412 (Elkhart Lake) on SPC-10W35.
- Ubuntu/Linux: Not explicitly confirmed for SPC series. [unverified]
- Modularity: 204-pin DDR3L SODIMM (up to 8GB) — RAM serviceable; CPU soldered/BGA (not swappable). M.2 WiFi support.
- Fanless: Yes, "Fanless Stainless Steel Chassis."
- VESA: Confirmed, VESA 75/100mm.

- Lifecycle: General product-life-cycle-management service line; for make-to-order items, "manufacturing can be reactivated after EOL based on demand." 2-year standard warranty (1 year ODM/OEM, 6 months for out-sourced parts kits).
- Power: 100-240VAC via 60W AC adapter (24V@2.5A) — notably AC-fed rather than direct DC.
- Also notable: LinuxGizmos-covered Avalue OFT-07W33 (7", Apollo Lake) explicitly ships with Android, Ubuntu, or Windows 10 as a factory OS choice — real evidence Avalue treats Ubuntu as first-class on at least some open-frame SKUs, with larger 15.6"/21.5" Apollo Lake variants noted too. <https://linuxgizmos.com/open-frame-touch-panel-offers-android-and-ubuntu-options/>

2.2.10 Axiomtek (GOT series)

- Models: GOT321W-521 (21.5", LGA1151 8th/9th Gen Core — genuinely socketed), GOT321B-ADL-WCD (21.5", Alder Lake), GOT3217W-881-PCT (21.5", discontinued, replaced by GOT3217WL-845-PCT), ITC241 (23.8", modular, Intel Smart Display Module architecture).
- CPU: 4th-9th Gen Core (several genuinely LGA-socketed, a rarity in this category) plus Alder Lake on newer models. ITC241 uses swappable SDM cards (Pentium N4200/Celeron N3350/4305UE/Core options).
- Ubuntu/Linux: Not explicitly confirmed. [unverified]
- Pricing: Available via Mouser Electronics — one of the more accessibly-purchasable vendors (not pure B2B quote-only).
- Modularity: Best-in-class among "standard" vendors — several GOT models use genuinely socketed LGA CPUs (field-replaceable/upgradeable processor, rare in this category), plus the ITC241's SDM swappable compute card. Dual mini-PCIe, 2.5" SATA bay, CFast socket.
- Fanless: Yes.
- Touch: PCAP confirmed ("PCT" suffix = Projected Capacitive Touch), 10-point on ITC241.
- VESA: Explicitly confirmed — "panel mounted, wall mounted, VESA mounted, or desk mounted."
- Lifecycle: No explicit dated statement, though LGA-socket approach inherently extends field-serviceable life.
- URLs: <https://www.axiomtek.com/Default.aspx?MenuId=Products&FunctionId=ProductView&ItemId=26930&upcat=375> (ITC241)

2.2.11 Kontron (FlatClient / KPanel)

- Models: FlatClient ECO-AML/ADN (Atom X7000E / Intel N-series / Core i3-N), FlatClient PRO-RPL (13th Gen Raptor Lake, 10-core), KPanel S-RPL. Sizes span 10.1"-23.8".
- CPU: ECO line includes N-series (N100-class); PRO-RPL is 13th Gen — among the most current CPU generations found across all 13 vendors.

- Ubuntu/Linux: Supports Windows 7/8.1/10 IoT and "Linux" generically; also offers proprietary KontronOS (hardened Linux-based OS for Flat-Client), which is not stock Ubuntu — needs direct confirmation whether vanilla Ubuntu LTS is separately supported if the studio's toolchain needs standard Ubuntu package compatibility. [ambiguous]
- Modularity: Not explicitly detailed; described as "closed, fanless metal housing," suggesting a more sealed/integrated design than AAEON/Cincoze/DFI.
- Fanless: Yes.
- Touch: Both PCAP and resistive options offered.
- Lifecycle — strong, explicit: "Long-term supply program with availability of 10 years or more." Related rail-grade product (Kontron TRACe HMI) stated at "20 years or more." VX30101 (different product line): "planned availability of at least 10 years," plus a dedicated long-term-services package beyond EOL and up to 5-year warranty extension. Kontron maintains named "(EOL, extended lifecycle program)" product pages as ongoing practice — one of the more mature EOL communication practices surveyed.

2.2.12 ASUS IoT

- Models: APC-125U series (2025) — APC-125U-15S, -17S, -215W, covering 15", 17", 21.5".
- CPU: Intel Core Ultra 5-125U (Meteor Lake) with Arc Xe-LPG integrated graphics and NPU — one of the most current/capable GPU/CPU combos of any vendor surveyed, notable for real-time graphics (Arc Xe-LPG's compute-shader/OpenCL performance meaningfully exceeds older UHD graphics on N100/Alder-Lake-N parts).
- Ubuntu/Linux: APC-125U description mentions "Industrial Android & Linux FOTA" support; ASUS IoT has a formal Canonical partnership demonstrated via the PE100A ("first NXP i.MX8 device certified by Canonical for Ubuntu 20.04," up to 10 years Linux security, 5 years bundled ESM) — but PE100A is ARM, not x86, so this certification does not directly transfer to APC-125U. [inferred partial transfer, flagged uncertain]
- Modularity: SO-DIMM DDR5 (user-serviceable, up to 32GB), two M.2 slots (B-Key + E-Key) for modular expansion. CPU is soldered SoC (not swappable).
- Fanless: Yes, ~50mm deep chassis.
- Touch: PCAP confirmed.
- Lifecycle — precisely quantified: "5-year supply guarantee" for motherboards, extendable by negotiated purchase order for >5yr projects. Named multi-stage EOL process: ECN before revisions → EOL notice 1 year before EOL → Last Buy Order 6 months before EOL — the most granular, multi-milestone EOL communication process found in this entire survey.

2.2.13 Shuttle

- Models — most current CPU generation lineup surveyed: P15EL01 (15.6", Elkhart Lake, fanless), P15RL01 (15.6", 13th Gen Raptor Lake-U, 15W TDP), P15AR01 (15.6", Arrow Lake-U), P21AR01 (21.5", Arrow Lake-U, AI Boost NPU, up to 64GB DDR5-6400).
 - CPU: Spans Elkhart Lake through Raptor Lake-U and Arrow Lake-U — genuinely 2024-2025 silicon, directly relevant to real-time graphics given Arrow Lake's improved iGPU.
 - Ubuntu/Linux: Shuttle maintains a dedicated "Linux Compatibility" support page listing which models are Linux-validated (specific P-series panel PC entries not confirmed in this pass — direct page review recommended: <https://global1.shuttle.com/Support/LinuxCompatibility>). General category language confirms "Microsoft Windows IoT, Ubuntu Linux, Yocto Embedded Linux... with Long-Term Support" as supported options.
 - Modularity: DDR5 SO-DIMM (dual-channel, serviceable, up to 64GB); CPU is soldered BGA mobile-U-series (not swappable).
 - Fanless: P15EL01 explicitly fanless; higher-TDP Raptor Lake-U/Arrow Lake-U configs need per-SKU confirmation.
 - VESA: Confirmed generally for the P-series category.
 - Lifecycle: No dedicated longevity program found. 3-year warranty on select barebone models (parts+labor, depot service); 1-year default otherwise.
-

2.3 2. ARM Panel PCs — Vendor-by-Vendor

2.3.1 Forlinx

- No dedicated turnkey panel-PC product line matching target sizes — primarily a SoM/SBC vendor (FET3568-C/OK3568-C = RK3568, Mali-G52; FET3588-C/OK3588-C = RK3588, quad A76+quad A55, 8K output). One low-end HMI tablet (FDU070S-R01) is actually i.MX RT1052 (Cortex-M7, no MMU/Linux capability) — not comparable.
- OS/kernel: OK3588-C has a Yocto 5.0/Linux Kernel 6.1 BSP and a separate "Forlinx Desktop 22.04" (Ubuntu-based) manual; older RK3588 docs reference kernel 5.10.209 — check per-product. New RK3572 SoM (June 2026) ships with a Linux 6.12 BSP, showing active movement toward newer kernels.
- Lifecycle: Explicitly markets 10-15 year longevity/last-time-buy commitments, reinforced by NXP's Product Longevity Program for i.MX-based lines — clearest for NXP parts, less explicit for Rockchip lines.
- Assessment: Best fit is building a custom panel around their SoM+carrier, not buying an off-the-shelf integrated unit.

- URLs: <https://www.forlinx.net/product/rk3588-som-134.html>, https://docs.forlinx.net/rockchip/ok3588-c/OK3588-C_Forlinx_Desktop22_04_User_Manual.html

2.3.2 Firefly

- AIO-3588Q — the closest match to a "build your own panel PC" board: RK3588, up to 32GB RAM, 8K decode, 5 display outputs (HDMI, DP, VGA, eDP, MIPI-DSI) for multi-screen, M.2 SATA/NVMe expansion. ~\$429-\$899 depending on RAM/storage config (marketplace-sourced, approximate). Not sold with an integrated touchscreen bezel — requires pairing with a separate panel.
- ROC-RK3588-PC / ROC-RK3588S-PC — RK3588(S), Mali-G610 MP4, 6 TOPS NPU, up to 32GB RAM.
- OS/kernel: Broadest official OS matrix found among ARM vendors — explicitly lists Android 12, Ubuntu (Desktop + Server), Debian 11, Buildroot, Kylin, UOS, with dedicated wiki build guides for both Ubuntu and Debian from source. https://wiki.t-firefly.com/en/ROC-RK3588S-PC/linux_compile_ubuntu.html
- Assessment: Strongest ARM candidate for a studio willing to pair a compute board with a separate touch panel; cheaper than most turnkey options.

2.3.3 Advantech (ARM line — PPC-100 series)

- PPC-115W (15.6") and PPC-112W (11.6") — both use RK3399 (dual A72+quad A53, Mali-T860) — notably older/lower-performance silicon (RK3399 launched ~2017) than RK3568/RK3588 competitors. 2-4GB LPDDR4 + 16-32GB eMMC (onboard/soldered). PPC-112W offers Debian OS option alongside Android, IP65-rated front, 10-point PCAP.
- Assessment: Real industrial build quality and genuine Debian support, but meaningfully behind on compute/GPU vs. newer RK3568/RK3588 vendors. No newer Advantech ARM refresh found in this pass — worth a direct RFQ to check.

2.3.4 AAEON (ARM line)

- ARM offerings are thin and don't reach target sizes: ACP-1078/1075 (7" RK3568 PCAP panel PCs) — below the 10.1" minimum. AAEON's actual 21.5" HMI answer, NIKY-2215-NX, uses NVIDIA Jetson Orin NX (Cortex-A78AE), not Rockchip — outside RK3568/RK3588 scope but a real alternative ARM path with NVIDIA's well-documented JetPack/Ubuntu support model.
- Assessment: Not currently competitive as an RK3588 panel-PC vendor at target sizes; Jetson-based NIKY line is the closest usable AAEON ARM option.

2.3.5 Boardcon

- No integrated panel PC — pure SoM/SBC vendor (EM3568, SBC3568, MINI3568) with separate 7"/10.1" LCD expansion modules. Touch HID compliance, OS image support, and lifecycle policy could not be confirmed. Lowest evidence confidence of all ARM vendors researched.

2.3.6 Geniatech — broadest RK3568/RK3588 size range found

- TPC-K3 series spans 10" to 32", RK3568-based, 10-point PCAP, Android 11.
- Specific SKUs found: 27" TPC2700 (~\$549), 15.6" TPC1560K4 (~\$469), 15.6" Android industrial panel PC (RK3566), 32" all-in-one (possibly older Cortex processor on the TPC3200 variant — verify SoC before buying).
- APC3568 (fanless ARM mini PC, RK3568, \$169-259) and APC3588 (octa-core RK3588, up to 32GB RAM, 8K encode/decode, \$269-359) are box-form compute, not integrated panels — usable as a modular pairing.
- OS: Vendor material states support for "Android 11.0 and Linux (Buildroot/Debian/Yocto)" — Debian explicitly named; Ubuntu not explicitly named for the panel-PC line itself. A third-party OEM using the same RK3568 chip (Panelmate) explicitly advertises a 15" panel PC running "Android and Ubuntu 20.04" — real evidence Ubuntu builds are achievable on this hardware category generally, though Ubuntu 20.04 is an aging LTS. [marketplace listing]
- RAM/storage: Search snippets suggest RK3588 Geniatech boards use "a standard SODIMM interface" (socketed) — unconfirmed whether this applies to the panel-PC line vs. just SoM/dev-board products.
- Assessment: Strongest match for exact size-range coverage (10.1"-32" all represented) among true panel-PC vendors; Debian is vendor-confirmed, Ubuntu is plausible but needs direct confirmation per-SKU.

2.3.7 Bonus finds (not on original list, relevant)

- TouchThink — real, size-complete RK3588 panel PC catalog: 10.1" (TPC101-A5), 11.6", 15"/15.6" (TPC156-A2, IP65), 17"/17.3", 19", 21.5" — all fanless aluminum chassis, IP65 front, PCAP touch, 6 TOPS NPU. Also a 10.1" RK3568 variant (TPC101-A2). OS support (Android vs. Linux) not explicitly stated — appears Android-first.
- FriendlyElec CM3588/CM3588 Plus (SoM+carrier, needs panel) — officially supports Debian 11, Ubuntu 22.04, Android 12, FriendlyWrt on Linux 6.1 LTS kernel — one of the most modern, clearly documented BSP base-lines found among any ARM vendor in this survey.
- Radxa (ROCK 5 ITX + Display 8 HD/Display 10 FHD panels) and Khadas (Edge2 + TS050, official Ubuntu firmware v230425+) — both maker/prototyping-tier, useful reference points for BSP maturity but not

turnkey industrial panel PCs, and Khadas tops out at 5" (below target range).

2.3.8 RK3588 Linux/GPU maturity — critical findings for the buying guide

- Mainline kernel status: As of end-2024, mainline Linux still lacked HDMI output on RK3588. Linux 6.10 added the Mali-G610 GPU driver (Panthor) without display support yet; 6.13 added HDMI; 6.14 (~March 2025) added MIPI-DSI. Collabora's own projection: mainline "should be usable" with 3D accel, some codecs, NPU by Q2 2025.
- GPU conformance: The open-source Panthor driver achieved OpenGL ES 3.1 Khronos conformance on Mali-G610 — a real, certified milestone.
- Video decode: As of February 2026, mainline gained H.264/H.265 hardware decode for RK3588's decoder cores (Collabora, 17-patch series), with GStreamer 1.28 support; multi-core decode, AV1, VP9 still future work.
- Vulkan caveat — load-bearing for a real-time graphics buying decision: real-world evidence (Orange Pi 5 forum/GitHub) shows OpenGL ES via EGL is the reliable, hardware-accelerated path on stock Ubuntu RK3588 images today; Vulkan windowed/surface rendering is NOT reliably supported outside headless compute use. This is a genuine gap vs. mature x86 Vulkan+Wayland/X11 support.
- Community distro risk: Joshua-Riek/ubuntu-rockchip, the most prominent community Ubuntu port for RK3588 (used by Rock 5B, Orange Pi 5, etc.), was archived/discontinued by its maintainer on April 29, 2026 — a significant red flag for 10-year serviceability planning, since the leading grassroots Ubuntu path for this SoC family has lost active maintenance. Arm-bian remains an active alternative.
- Demonstrated real-time graphics capability: A hobbyist project (Hackaday.io) renders "hundreds of Pac-Man/Ghost sprites at 60 FPS" and "thousands of rotating sprites at 60fps" via OpenGL ES 3.1 + Mesa Panfrost on Orange Pi 5 Ultra — concrete evidence Mali-G610 + open driver stack handles moderate 2D/sprite-based real-time work, but this is far short of a rigorous 3D/shader benchmark, and no TouchDesigner/openFrameworks-specific RK3588 benchmark was found anywhere in this research.
- Touch HID caveat: Many integrated ARM panel PCs (Forlinx, Geniatech, TouchThink, Firefly AIO boards) connect touch via I2C directly to the SoC, not USB — functionality then depends on the specific touch controller's kernel driver being present in the vendor BSP (or mainlined), not generic USB HID auto-detection. This should be confirmed per-model.

2.4 3. Modular "Replaceable Compute" Display Architectures: OPS and SDM

2.4.1 OPS (Open Pluggable Specification)

- Standard: Announced 2010 by NEC, Intel, Microsoft. Module envelope 180×119×30mm (200×119×30mm with mounting frame). Connector: single 80-pin JAE TX24/TX25 blind-mate board-to-board connector (1.27mm pitch, ~500 mating cycles). Power: +12V to +19V DC across up to 8 power pins (~8A max). Carries DVI-D/DisplayPort/HDMI video, USB 2.0/3.0, PCIe x1, stereo audio, RS-232. OPS+ adds a secondary high-speed connector.
- Displays with OPS slots: LG (CreateBoard interactive displays, OPSJ-5LDJA and OPS-C001 modules); Samsung (Flip 3/Flip Pro via CY-PBRK200XEN — note Samsung also has a separate, proprietary "PIM" slot that looks similar but is NOT the open OPS standard — don't conflate the two); NEC/Sharp ("all NEC Large Screen displays except E-Series feature the OPS option slot" — but NEC's newer P-series has moved to "OMI" slot, which takes SDM-S/SDM-L or Raspberry Pi CM4, not classic OPS — check per-model); ViewSonic CDE31 series (43"-98", OPS confirmed on 65"+ models, modules VPC37-W55-G1, VPCF5-W55-G1 with 12th Gen Core, VPC13-C33-G1 ChromeOS, VPC-A31-O1 Android); BenQ (75" 4K signage + first Chromebox OPS module TEY1C, Q2 2025); Philips/MMD (42BDL5057P and larger 75"/86"/98" units); Iiyama (OPC51103BC module with 10th Gen i5-1135G7, paired to a defined list of 55"-98" ProLite/interactive touchscreens).
- Notable non-adopters/migrators: Panasonic (current EQ2/SQ1/SQE2 lines use SDM, not OPS); Sony (BRAVIA Professional Displays use built-in SoC, no OPS slot — not a good modular-serviceability match); DynaScan (moved to SDM).
- OPS compute modules (x86): Advantech DS-280 (6th Gen Core i7/i5/i3, up to 3 displays at true 4K); Kontron OPS-1000/2000/2010/KOPS800 (4th-Gen through Skylake/Kaby Lake); IBASE IOPS-602 (7th Gen Core, up to 32GB DDR4 SODIMM, explicitly runs both Windows 10 and Linux Ubuntu — the single strongest direct Ubuntu evidence found for any OPS module); Axiomtek OPS500 series (6th-9th Gen, appears to lag current silicon); Giada PC612 (12th Gen Alder Lake, 8K support) and Giada F108D (Alder Lake-N, fanless, N100 variant — confirms modern low-power silicon is being packaged in OPS form factor as of 2025); JWIPC S104 (12th Gen); Gigabyte GB-SIOPS.
- OPS compute modules (ARM): Geniatech OPS3399 (RK3399, Android 10); JWIPC S088 (RK3588); legacy Samsung SBB-SSE (Cortex-A9) and ZidooLab (RK3368/RTD1295).
- Field replacement: Marketing calls OPS "hot-swappable," but official installation guides (SMART Technologies) require the display to be powered off, switch OFF, ≥30 seconds wait for internal supplies to discharge

before removal. Realistic characterization: tool-free, tray-and-screw field-replaceable in under 5 minutes with display powered off — not true hot-swap.

- Longevity assessment: OPS is a genuinely open, multi-vendor, ~15-year-old standard with modern silicon (Alder Lake-N, some Alder Lake-P) still shipping in the form factor as of 2025 — structurally favorable for long-term multi-source module replacement. Caveat: no vendor publishes a formal “we guarantee OPS module availability for X years” commitment, and several major display vendors are visibly migrating new flagship lines to SDM/OMI. Verify the exact slot type per specific model rather than assuming “OPS” branding is uniform across a vendor’s whole catalog.

2.4.2 Intel SDM (Smart Display Module) / SDM-S / SDM-L

- Standard: Announced ISE 2017 as a smaller successor/complement to OPS for ultra-slim AIO/interactive-panel designs. SDM-S = 60×100mm; SDM-L = 175×100mm, max 20mm z-height without enclosure (bare board, no OPS-style metal housing — this shrinks integration depth vs. OPS). Custom high-speed connector (not the OPS 80-pin JAE). Supports up to 8K.
- Modules: Giada SDM-L613 — current-generation, 13th Gen Intel Core i3-1315U/i5-1335U/i5-1345U/i7-1355U (Raptor Lake-U), dual-channel DDR5-5200 up to 64GB, Iris Xe graphics, HDMI 2.1 up to 8K — strong evidence SDM is receiving genuinely current silicon. Also Axiomtek SDM500L, GigaIPC SDM-1185G7EL. Advantech offers modular signage solutions certified for both SDM and OPS.
- Displays: NEC/Sharp OMI-slot displays (P654Q, M861 IGB) accept SDM-S/SDM-L; Panasonic EQ2/SQ1/SQE2; DynaScan.
- Maturity vs. OPS: SDM is newer and actively developed (Intel published an “SDM Partners Catalogue 1H2024”), but has a materially smaller multi-vendor module marketplace than OPS — roughly 4 module makers found (Giada, Axiomtek, GigaIPC, Advantech) vs. 9+ for OPS. This is a real risk for a 10-year replaceability strategy: fewer independent suppliers means less redundancy if one discontinues a line.
- Linux/Ubuntu: Marketing language states SDM modules “support Windows, Linux and Android 5.1” generically — no specific Ubuntu-version certification found (weaker evidence than IBASE’s explicit OPS/Ubuntu claim). Because current SDM modules use mainstream Intel Raptor Lake-U silicon with mature mainline Linux driver support, Ubuntu compatibility is plausible/likely in practice even without an explicit vendor certification statement. [inferred]

Bottom line for the guide: OPS remains the safer bet today for breadth of module sourcing and proven multi-vendor competition; SDM is real and getting current silicon but is a thinner ecosystem. Both are viable “swap the brain, keep the glass” strategies — arguably a better longevity architecture than any sin-

gle sealed panel PC, since the display and compute genuinely fail/obsolete on different timelines and OPS/SDM lets you replace only what needs replacing.

2.5 4. Product Lifecycle & Warranty — Cross-Vendor Comparison

Vendor	Longevity claim
Advantech	"10+ years" platform; 5-10yr roadmap guarantee
AAEON	"3-5 years or longer" (Intel roadmap)
OnLogic	Named: "guaranteed until 2035" (Factor 101)
Cincoze	General "10-15yr industry" framing only
Teguar	"Exact same configuration for life of project"; 10+ yrs, tied to Intel Embedded Roadmap
Winmate	Product Longevity Program: 5-yr stable supply
DFI	"Up to 15-year" longevity; named example to Q3 2035 (EC70A-TGU)
IBASE	Not found
Avalue	General PLM service; EOL "reactivable" for MTO
Axiomtek	General "long-term availability" only
Kontron	"At least 10 years"; SBCs "up to 10 years"; rail HMI "20 years+"
ASUS IoT	"5-year supply guarantee," negotiable beyond
Shuttle	Not found
Crystal Group (adjacent)	Configuration-management framing

Intel's underlying embedded processor longevity program (the mechanism behind most vendor claims):

- "Embedded Options Available" SKUs are typically available 7 years from launch of the first SKU in the family, possibly longer, but this is explicitly not a contractual guarantee — Intel reserves the right to change roadmaps and discontinue via standard PCN/PDN process. [vendor page]
- Elkhart Lake (Atom x6000E / Pentium/Celeron N/J-series, Q1 2021) is cited by secondary module-vendor sources (TQ-Group, congatec) as carrying "guaranteed availability of 15 years" — this is a vendor-secondary characterization, not directly confirmed against an Intel first-party page in this research pass.
- N97/N100 (Alder Lake-N, used in many current fanless panel PCs): these are consumer-branded "Intel Processor N-series," and it is not confirmed they carry the same "Embedded Options Available" long-life SKU designation as Elkhart-Lake-class or "E"-suffix embedded parts. One vendor motherboard datasheet claims "10-year longevity through 2034" for N-series, but this is unverified against Intel's own ARK data. Recommendation: check Intel ARK's "Embedded Options Available" flag

for the exact SKU before relying on any vendor's longevity marketing for N97/N100/N305 parts.

Key takeaway: Only OnLogic and DFI were found with concrete, dated, SKU-specific commitments ("guaranteed until 2035" / "available until Q3 2035") rather than generic "X years" marketing language — these are the strongest citable examples. ASUS IoT has the most transparent multi-stage EOL communication process. Crystal Group (rugged/adjacent category) has the longest warranty stack if the studio wants single-vendor coverage spanning the full 10-year horizon rather than relying on post-EOL spares promises.

2.6 5. Real-World Linux Evidence

2.6.1 PCAP touch = standard USB HID — the clearest finding of this research

- Elo Touch Solutions' own support documentation states unambiguously: "HID support was integrated into Linux kernel version 3.8, and if your Linux kernel version is 3.8 or greater, then there is no need for an Elo driver to enable touch features on a PCAP screen... Elo's PCAP and IR products are completely HID compliant and typically do not require a touch driver to be installed." [vendor support page, high credibility] <https://elosupport.elotouch.com/hc/en-us/articles/31649725511447>
- This matches the mainline kernel's `hid-multitouch.c` driver, auto-binding to any USB device presenting a standard HID multitouch descriptor. ArchWiki corroborates: "for many devices, support works automatically without needing to install additional drivers."
- Real exceptions found: (1) Older/legacy Advantech resistive serial touch panels do NOT work out of the box — require manual `inputattach` + `systemd` service + `xinput-calibrator` (this is resistive/serial, not PCAP/USB, so a different technology, not a contradiction). (2) Cincoze/OnLogic's own official documentation shows at least some of their touch panels require a manual eGalax driver install (`setup.sh`, select "USB" mode) rather than relying on generic HID — a genuine counter-example to "always plug-and-play," and OnLogic explicitly warns touch will not work during initial OS install, so plan to have a keyboard/mouse on hand during commissioning.
- Bottom line for the guide: On stock Ubuntu Desktop x86_64 with a genuinely HID-compliant PCAP panel, zero driver install should be needed — but verify per-vendor/SKU (check via `xinput list / dmesg | grep -i hid` after connecting) and ask for a stock (not custom BSP) Ubuntu image where possible. ARM panel PCs are a different story — many connect touch via I2C directly to the SoC rather than USB, making functionality dependent on the vendor BSP's specific touch-controller driver rather than generic HID auto-detection.

2.6.2 N100/N97/Alder-Lake-N specific Linux gotchas

- Multiple independent forum threads (Linux Mint Forums, Arch Linux Forums) converge: "Alder Lake N-Refresh" processors (N97, N150, N250, Core i3-N355) need kernel ≥ 6.11 for the i915 graphics driver to load. Plain original N100 is fine on older kernels; it's specifically the newer "N-Refresh" SKUs that need very recent kernels. This means Ubuntu 22.04 LTS (or even 24.04 at GA, kernel 6.8) may need an HWE kernel upgrade — a real, actionable spec-sheet checklist item.
- CNX Software/Linuxium's hands-on Beelink SEi12 (Core i5-1235U, Alder Lake-P — same graphics generation) review under Ubuntu 22.04.1 found: unresolved but non-fatal ACPI/MSR kernel warnings (open Launchpad bug), and a genuine dual-4K-display regression on Ubuntu that doesn't exist on Windows on this exact hardware — relevant if the studio wants dual-output panel PC setups. Audio/WiFi/Bluetooth/Ethernet all worked fine.
- N100 media decode: solid for standard 4K60 (AV1/H.265 hardware decode), but a Kodi forum thread found 8K AV1 pushes N100 to its limit, with a second concurrent stream becoming "severely sluggish" — a real ceiling if the studio plans very high-res concurrent video playback.

2.6.3 Media-art / TouchDesigner / openFrameworks community hardware reality

- TouchDesigner forum consensus (multiple threads, 2017-2024, recurring roughly annually): the community explicitly frames Intel integrated graphics — any generation — as adequate only for "noodling around," not "intensive graphics or tasks." Discrete NVIDIA GPU (Quadro preferred for permanent installs due to tear-free multi-display sync; GeForce acceptable for smaller gigs) remains the de facto standard recommendation for serious real-time work. <https://forum.derivative.ca/t/does-anyone-have-experience-with-intel-nucs/208491>
- openFrameworks forum (2014, revisited 2019): a user building a 24/7 12-month touchscreen install explicitly rejected Intel NUCs due to "pretty bad reviews," using an alternative mini-PC instead.
- Zero mentions of Advantech, AAEON, Cincoze, Winmate, or IBASE were found in any searched TouchDesigner or openFrameworks forum thread. The media-art community's default hardware vocabulary is consumer/prosumer mini-PCs (NUC, Zotac with GeForce cards, Beelink, ACEMAGICIAN) or custom-built Mini-ITX rigs with discrete AMD/NVIDIA GPUs (e.g., Function Store's Patreon post on building custom Mini-ITX PCs specifically for TD installations rather than buying off-the-shelf hardware). This means the studio would have essentially no peer community troubleshooting history to draw on if adopting industrial panel PC brands — budget extra validation time.
- General (non-TD-specific) benchmark data: Iris Xe (12th/13th gen U-

series) is roughly 2-4x faster than older UHD Graphics in 3DMark/gaming benchmarks; N100's UHD (24 EU) scored a very low 258 points/10.2 avg fps in Unigine Heaven 4.0 @1080p — confirms N100/N97-class iGPUs are not suited to demanding real-time 3D/shader work by any conventional benchmark standard. No direct TouchDesigner- or openFrameworks-specific N100/N97 benchmark exists in any indexed source — this is a real gap; the studio would be establishing new ground truth, not confirming existing community data.

- A blunt counterpoint worth including: a 15-year industrial-automation veteran on the LinuxCNC forum argued that "industrial" panel PCs often just mean "a rugged metal case around otherwise mediocre/dated internals," and that a repurposed commercial PC (e.g., a Dell 9020 + separate Dell touch monitor) can dramatically outperform an "industrial" unit for a fraction of the cost — a real risk to weigh against the serviceability/longevity benefits documented above.

2.7 6. GPU/Real-Time Graphics Comparison: N100-class vs Core-class vs RK3588

Tier	Representative chip
N100-class (Alder Lake-N)	Intel N97, N100, N105
Core-class (12th/13th Gen U-series, Iris Xe / Meteor Lake Arc Xe-LPG / Arrow Lake-U)	Core i5-1235U (Iris Xe), Core i7-13700H (Arc Xe-LPG)
RK3588 (Mali-G610 MP4)	Rockchip RK3588/3589

Practical recommendation: For genuine openFrameworks/TouchDesigner-class real-time graphics work (shader-heavy, particle systems, video-heavy compositing), none of the sealed industrial panel PC options are ideal — the studio should either (a) pick a panel PC with the strongest available iGPU (ASUS IoT APC-125U/Arc Xe-LPG or Shuttle Arrow Lake-U) for lighter workloads, or (b) use an OPS/SDM-slot commercial display paired with a compute module that has a discrete GPU option, or (c) follow the TD community's own well-worn path of a separate discrete-GPU Mini-ITX PC driving an industrial touch monitor (not a panel PC with integrated compute) — sacrificing the single-enclosure convenience for real GPU headroom and a field-swappable compute box.

2.8 7. Sources Index (key URLs by category)

x86 vendors: advantech.com/en/products/ubuntu-lts-os, ubuntu.com/certified/vendors/Advantech, aaeon.com/en/p/omni-series-modules-panels, onlogic.com/store/computers/panel-pc/tacton, ubuntu.com/certified/202304-31510, cincoze.com/en/goods_catalog.php?cid=2,

catalog.redhat.com/hardware/system/detail/134917, teguar.com/tp5010-long-life-cycle-panel-pcs, winmate.com/en/Product/W27IAD7T-PCA1, dfi.com, cnx-software.com/2022/12/05/dfi-ec70a-tgu-embedded-computer-long-term-support-2035, ibase.com.tw (2025-2026 catalog PDF), avalue.com/en/product/Panel-PC-Monitor-Mobile, axiomtek.com (ITC241), mouser.com/manufacturer/axiomtek, kontron.com/en/products/systems/panel-pc/flatclient, iot.asus.com/us/resources/news/apc-125-series, global1.shuttle.com/Support/LinuxCompatibility

ARM vendors: forlinx.net/product/rk3588-som-134.html, docs.forlinx.net, en.t-firefly.com/product/industry/aio3588q, wiki.t-firefly.com/en/ROC-RK3588S-PC, advdownload.advantech.com (PPC-115W/112W datasheets), cnx-software.com (NIKY-2215-NX, RK3588 mainline status articles), board-con.com/EM3568_SBC, geniotech.com/products/custom-panel-pc, touch-think.net, friendlyelec.com, radxa.com/products/rock5/5itx, khadas.com

OPS/SDM: en.wikipedia.org/wiki/Open_Pluggable_Specification, sharp-necdisplays.us/ops, viewsonic.com/business/cde31-series, digitalsignageto-day.com (IBASE IOPS-602), giadatech.com (Alder Lake OPS/SDM news), intel.com/content/www/us/en/products/docs/smart-display-module/overview.html, downloads.smarttech.com (OPS install guide)

Lifecycle: advantech.com/DMS/Services/Longevity-Services, onlogic.com/company/io-hub/why-product-lifecycle-matters, teguar.com/choosing-a-cpu-for-industrial-computers, kontron.com/en/media/news/kontron-launches-vx30101, asus.com/us/content/5-year-longevity, crystalugged.com/support-team/warranty, intel.com/content/www/us/en/support/articles/0000 (Embedded Options Available)

Real-world Linux/TD evidence: elosupport.elotouch.com (PCAP HID), support.onlogic.com (touch driver install), cnx-software.com/2024/07/08 (Cincoze DS-1402 Ubuntu 24.04 review), forum.derivative.ca (TouchDesigner NUC/PC-specs threads), forum.openframeworks.cc, forum.linuxcnc.org/18-computer/46152-industrial-panel

2.9 Evidence quality legend

- [datasheet] — directly fetched vendor PDF/spec sheet
- [vendor page] — directly fetched or high-confidence vendor webpage
- [vendor page — via search snippet] — vendor content, but only seen via search result excerpt (web_fetch was rate-limited for large portions of this research); treat as needing final direct verification
- [distributor page] — Mouser/Digi-Key/Newark/ProVantage etc.
- [forum] — user/community forum post, reddit, or independent blog/review
- [press] — trade press (CNX Software, LinuxGizmos, etc.)
- [inferred] — reasonable conclusion drawn from adjacent evidence, not directly stated

- [unverified] — claim found but could not be corroborated; flag for direct vendor confirmation before purchase decision